Predicting the Predictable: Decomposing and Forecasting Stock Returns in a Data-rich Environment *

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Abstract

We revisit aggregate stock return predictability from a target-design perspective. The difficulty of beating the historical mean may reflect not only weak predictors or inadequate algorithms, but also the fact that aggregate returns pool components with different signal-to-noise ratios. We propose a factor-augmented sum-of-the-parts (FA-SOP) framework that directs high-dimensional macroeconomic information to valuation-multiple growth, the component where present-value logic suggests macroeconomic information should be most visible, while retaining parsimonious forecasts for lower-signal components. Applied to S&P 500 returns from 1960 to 2022, FA-SOP delivers robust out-of-sample statistical and economic gains over the historical mean, univariate regressions, direct factor-augmented return regressions, and traditional SOP benchmarks. The key comparison is that the same underlying macroeconomic panel performs poorly when used to forecast aggregate returns directly, but becomes useful when routed through valuation adjustment. Machine learning extensions show that this pattern is not specific to the linear factor specification, and simulations with known expected returns confirm that FA-SOP better recovers time-varying expected returns without excessive forecast variance. The results support a broader "predicting the predictable" principle for data-rich empirical asset pricing.

Keywords: Return predictability; Return decomposition; Macroeconomic factors; Factor-augmented regression; Machine learning.

JEL Codes: C58; G12; G17.

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1 Introduction

Whether stock market returns are predictable remains a central question in empirical asset pricing. Expected market returns may vary with discount rates, cash-flow expectations, risk aversion, and macroeconomic conditions, and this possibility has motivated a large literature on aggregate time-series return predictability using valuation ratios, macroeconomic variables, financial conditions, forecast combinations, and high-dimensional factor methods (e.g., Campbell, 1987; Campbell and Shiller, 1988a; Fama and French, 1988, 1989; Hodrick, 1992; Kothari and Shanken, 1997; Lettau and Ludvigson, 2001; Lewellen, 2004; Ang and Bekaert, 2007; Cochrane, 2008; Ludvigson and Ng, 2007; Rapach et al., 2010; Giovannelli et al., 2021). Yet the out-of-sample evidence remains mixed. Many proposed predictors fail to beat the historical mean in the original Welch and Goyal (2008) evidence and in the more recent update of Goyal et al. (2024). The historical mean therefore remains a demanding benchmark, not because it is economically rich, but because direct return forecasts often add more estimation error than useful signal (Campbell and Thompson, 2008).

This paper argues that part of this difficulty is a target-design problem. Aggregate returns pool components with different economic content, persistence, and signal-to-noise ratios. A direct return regression asks the same predictor set to explain both the component where predictable variation may be concentrated and the components dominated by short-horizon noise. In a data-rich environment, adding more predictors or more flexible algorithms can therefore amplify estimation variance without improving the measurement of expected returns. We study a simple alternative principle: "predicting the predictable." Forecasting should be organized around the component where economic theory suggests that the signal is most likely to appear, and richer information should be directed there rather than applied mechanically to aggregate returns.

Our starting point is the Sum-of-the-Parts (SOP) framework of Ferreira and Santa-Clara (2011) (hereafter FSC). SOP decomposes the aggregate log return into

$$r_{t+1} = dp_{t+1} + ge_{t+1} + gm_{t+1}, \tag{1}$$

where r_{t+1} is the aggregate log stock return, dp_{t+1} is the dividend-price component, ge_{t+1} is earnings growth, and gm_{t+1} is price-earnings multiple growth. This decomposition is useful because it lowers estimation error relative to unrestricted predictive regressions. It also cre-

ates a sharper question about where predictability should be searched. The *gm* component is the change in the market's valuation of a given earnings stream; it captures valuation adjustment rather than realized cash-flow growth. Because valuation multiples summarize investors' forward-looking discounting of future cash flows, *gm* is a natural margin through which changing discount rates, risk appetite, and macro-financial conditions can affect short-horizon returns.

FSC also reveal the central tension that motivates this paper. Their passive SOP rule sets expected multiple growth to zero and forecasts the other two components parsimoniously. They also consider an active reversion specification in which the price-earnings multiple reverts toward a conditional valuation level estimated from a single observable predictor. In their own evidence, however, this active step does not improve much on the passive benchmark: the average out-of-sample R^2 of the reversion specifications is about 1.18%, compared with 1.32% for passive SOP (Tables 2 and 3 in FSC). This result can be read as evidence that decomposition works mainly as a variance-reduction device and that active modeling of valuation adjustment adds little. We offer a different interpretation. The weak incremental performance may reflect imprecise measurement of the conditional valuation level, not the absence of predictable variation in *gm*.

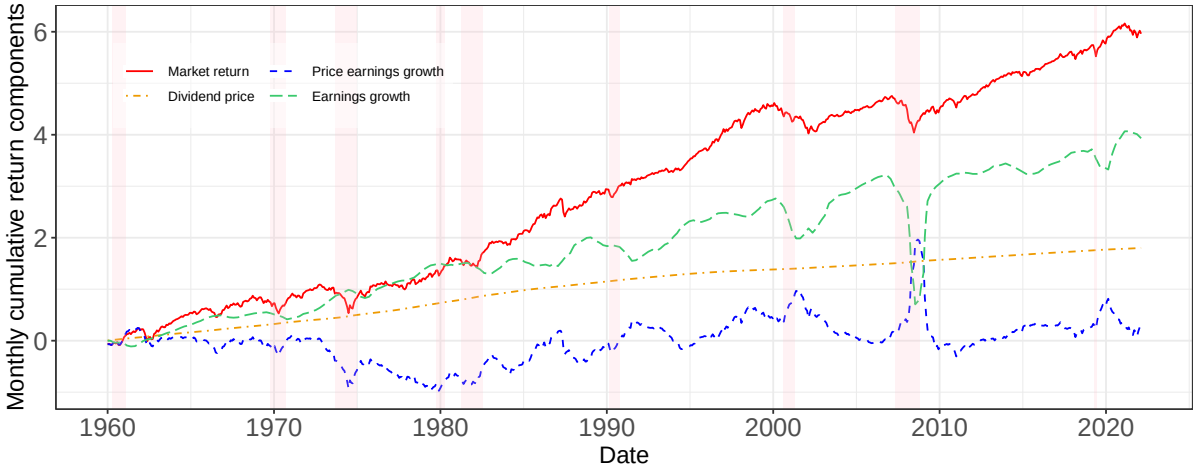
The economic motivation for focusing on valuation adjustment follows from the Campbell–Shiller present-value relation (Campbell and Shiller, 1988a). A compact implication of this relation, derived in Section 2.2, expresses the price-earnings multiple as a discounted sum of expected future earnings growth, expected future returns, and expected payout policy:

$$pe_t = \frac{\kappa}{1-\rho} + \sum_{j=1}^{\infty} \rho^{j-1} E_t(\Delta e_{t+j} - r_{t+j}) + (1-\rho) \sum_{j=1}^{\infty} \rho^{j-1} E_t(de_{t+j}). \tag{2}$$

Here, pe_t is the log price-earnings multiple, Δe_{t+j} is log earnings growth, r_{t+j} is the log stock return, de_{t+j} is the log dividend-payout ratio, and $E_t(\cdot)$ denotes the expectation conditional on information available at time t. The constants κ and $\rho \in (0,1)$ come from the standard log-linearization. Equation (2) shows why modeling the valuation multiple is not an ad hoc target choice. A macroeconomic state variable may have only a weak direct relation with next-period realized returns, because realized returns contain substantial transitory noise. The same state variable can nevertheless affect pe_t through expectations over many future horizons. A forecast of gm_{t+1} based on the deviation of pe_t from its macro-implied valuation level therefore uses macroeconomic information at the point where the present-value relation suggests it should be most visible.

Figure 1 provides a descriptive view of the decomposition. The dividend-price component is highly persistent and slow moving. Earnings growth varies more than the dividend-price component and displays an important low-frequency trend, but short-horizon earnings growth is difficult to forecast reliably (Campbell and Shiller, 1988b; Fama and French, 2002; Cochrane, 2008; Van Binsbergen and Koijen, 2010). Price-earnings multiple growth, by contrast, accounts for a large share of short-run valuation fluctuations. These patterns support a selective modeling strategy: retain parsimonious forecasts for dp and ge, following FSC's current-value and long-moving-average rules, but actively model gm through the adjustment of the price-earnings multiple toward a macro-implied valuation level.

Figure 1: Cumulative Return and Its Component Decomposition



Note: This figure shows monthly realized price-earnings multiple growth (gm), earnings growth (ge), the dividend-price component (dp), and stock market return (r). The vertical shaded area represents the business-cycle recession dates reported by the National Bureau of Economic Research (NBER).

We implement this idea through a Factor-Augmented Sum-of-the-Parts (FA-SOP) framework. FA-SOP preserves the low-estimation-error discipline of SOP for dp and ge, but replaces the passive treatment of multiple growth with a factor-augmented valuation-adjustment step. We extract latent macroeconomic factors from large-scale datasets such as FRED-MD and FRED-QD (McCracken and Ng, 2016, 2021) and use these factors, together with standard observable predictors, to estimate the macro-implied valuation level for the price-earnings multiple. The deviation between the observed multiple and this fitted level is an error-correction signal: when the observed multiple is below its macro-implied level, subsequent multiple growth is expected to be higher, and conversely when it is above that level.

Two design choices make FA-SOP a target-design framework rather than only a larger predictor model. First, high-dimensional macroeconomic information is directed specifically to

gm, instead of being used to estimate active forecasting equations for all return components. Second, the factor construction is matched to the time-series properties of the target. The price-earnings multiple is a persistent valuation object and is close to a unit-root process in our sample; gm_{t+1} is the change in that object. FA-SOP therefore uses level information in the macroeconomic panel and allows both stationary and nonstationary latent factors to characterize the macro-implied valuation level, consistent with the generalized factor structure of Hannadige et al. (2024).

The empirical design is organized around comparisons that separate information content from target organization. Using S&P 500 data from 1960 to 2022 at monthly and quarterly frequencies, we compare FA-SOP with four benchmark groups. The historical mean and univariate predictive regressions provide standard return-predictability benchmarks. FA-Unified uses the same underlying macroeconomic panel as FA-SOP but forecasts aggregate returns directly, allowing us to ask whether the same information is more useful when routed through valuation adjustment. Simple SOP and FSC-style SOP reversion hold the decomposition fixed while varying how the conditional valuation level is measured, allowing us to ask whether the weak performance of single-predictor reversion reflects the limits of active gm modeling or the limits of low-dimensional measurement.

The results support the target-design interpretation. FA-Unified performs poorly out of sample despite using the same broad macroeconomic information set. FA-SOP, by contrast, delivers positive and statistically significant out-of-sample gains. At the monthly frequency, FA-SOP reaches an out-of-sample R^2 of 1.95%, compared with 0.79% for the best FSC-style SOP reversion specification. At the quarterly frequency, FA-SOP reaches 6.09%, compared with 3.31% for the best FSC-style SOP reversion specification. These gains indicate that macroeconomic information is weak when used to forecast aggregate returns directly, but useful when directed toward valuation-multiple adjustment.

The economic value of the forecasts is also substantial, though with an important distinction across horizons. At the monthly frequency, FA-SOP generates larger certainty-equivalent return and Sharpe-ratio gains than the competing models, including after transaction costs. At the quarterly frequency, FA-SOP continues to deliver strong statistical performance, but its portfolio gains are closer to those of smoother SOP specifications. This pattern is consistent with a portfolio-implementation channel: more responsive forecasts improve conditional-mean tracking, but may also induce larger changes in portfolio weights and greater exposure to in-

terim reversals at lower rebalancing frequencies. We therefore interpret the economic results as supportive of FA-SOP, while also emphasizing that statistical predictability and implementable portfolio value need not move one-for-one.

We then examine why the comparison works. The evidence points to three related mechanisms. First, valuation adjustment is the main forecasting margin: a macro-implied valuation level for the price-earnings multiple generates an error-correction signal that predicts subsequent multiple growth more quickly than single-predictor reversion. Second, measurement of that level matters: latent macroeconomic factors summarize broad common variation in the data and cannot be replicated by a small set of hand-selected observable proxies. Third, the principle extends beyond the baseline linear specification. When commonly used machine learning methods are used to forecast aggregate returns directly, their out-of-sample performance is weak. When the same algorithms are embedded in the SOP structure and directed toward the valuation component, performance improves materially. This comparison suggests that the gains are not simply a consequence of a particular linear factor model; they arise from organizing the forecasting target before applying richer statistical methods.

Finally, we use a factor-augmented present-value simulation as a controlled diagnostic. The simulation generates returns, earnings growth, payout policy, and valuation multiples from macroeconomic state variables, so the true expected return is known. It reproduces the empirical pattern: direct return regressions have difficulty recovering expected-return variation because the single-period return signal is weak relative to realized return noise, while FA-SOP better tracks the true expected return by using macroeconomic information through valuation adjustment. The simulation therefore provides structural support for the interpretation that decomposition helps recover expected-return variation by targeting the component in which the macroeconomic signal is more concentrated.

The paper contributes to the return-predictability literature in two ways. First, it provides an implementable forecasting framework that combines the low-estimation-error discipline of SOP with high-dimensional macroeconomic information. Second, it clarifies a broader principle for data-rich empirical asset pricing: forecast performance depends not only on the information set or algorithm, but also on whether the forecasting target is economically organized. More reliable estimates of expected market returns can also serve as inputs for portfolio allocation, valuation, cost-of-capital estimation, and conditional asset-pricing analysis. Online Appendix H illustrates this point using expected market premia, industry cost of equity, and conditional-

CAPM diagnostics.

The remainder of this paper is organized as follows. Section 2 presents the return decomposition, the present-value motivation, the passive SOP restrictions, the FA-SOP implementation, the estimation procedure, and the evaluation criteria. Section 3 describes the data. Section 4 reports the empirical results, including the estimated latent macroeconomic factors and the statistical and economic performance of FA-SOP relative to benchmark models. Section 5 investigates the sources of predictability, focusing on valuation multiple reversion, latent macroeconomic factors, and the broader "predicting the predictable" principle beyond linear models. Section 6 conducts a Monte Carlo simulation based on a factor-augmented present-value model. Section 7 provides robustness checks across business-cycle states and alternative predictor sets. Finally, Section 8 concludes. Additional technical details, supplementary empirical results, simple-return and equity-premium forecasts, and the cost-of-equity and conditional-CAPM validation exercise are reported in the Online Appendix.

2 Methodology

This section develops the Factor-Augmented Sum-of-the-Parts (FA-SOP) framework. The goal is not simply to add predictors to a return regression, but to organize the forecasting target before adding information. We begin with the return decomposition and the present-value motivation for focusing on valuation adjustment. We then present the passive SOP restrictions, introduce FA-SOP as a factor-augmented valuation-adjustment method, and discuss how it preserves the low-estimation-error discipline of SOP while actively modeling the component where macroeconomic information is most likely to matter.

2.1 Return decomposition

Following Ferreira and Santa-Clara (2011), we decompose the log return into three accounting components. Let P_t denote the stock price, D_t dividends, E_t earnings, and $PE_t = P_t/E_t$ the price-earnings multiple. The gross return can be written as

$$\begin{aligned} 1 + R_{t+1} &= \frac{P_{t+1} + D_{t+1}}{P_t} \\ &= \frac{PE_{t+1}}{PE_t}\frac{E_{t+1}}{E_t}\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) \\ &= (1 + GM_{t+1})(1 + GE_{t+1})(1 + DP_{t+1}), \end{aligned} \tag{3}$$

where GM_{t+1} is price-earnings multiple growth, GE_{t+1} is earnings growth, and $DP_{t+1} = D_{t+1}/P_{t+1}$ is the dividend-price ratio. Taking logs gives the sum-of-the-parts identity

$$r_{t+1} = gm_{t+1} + ge_{t+1} + dp_{t+1}, \tag{4}$$

where lowercase variables denote log gross rates; in particular, $dp_{t+1} = \log(1 + DP_{t+1})$.

2.2 Present-value motivation for valuation adjustment

The decomposition above is an accounting identity. To motivate which component should be modeled actively, we use the log-linear present-value relation of Campbell and Shiller (1988a). Let pd_t denote the log price-dividend ratio, pe_t the log price-earnings multiple, $de_t = d_t - e_t$ the log dividend-payout ratio, and Δe_{t+1} and Δd_{t+1} log earnings and dividend growth. The Campbell–Shiller log-linear return approximation can be written as

$$r_{t+1} \approx \kappa + \rho pd_{t+1} + \Delta d_{t+1} - pd_t = \kappa + \rho pe_{t+1} - pe_t + (1-\rho)de_{t+1} + \Delta e_{t+1}, \tag{5}$$

where $\kappa = \log(1 + \exp(\overline{pd})) - \rho\overline{pd}$, $\rho = \exp(\overline{pd})/(1 + \exp(\overline{pd}))$, and $\overline{pd} = E(pd_t)$. The useful feature of this approximation is that it forms a forward recursion in the valuation ratio. Rearranging equation (5) gives

$$pe_t = \kappa + \rho pe_{t+1} + (1-\rho)de_{t+1} + \Delta e_{t+1} - r_{t+1}. \tag{6}$$

Iterating forward and imposing the transversality condition $\lim_{j\to\infty} \rho^j pe_{t+j} = 0$ gives

$$pe_t = \frac{\kappa}{1-\rho} + \sum_{j=1}^{\infty} \rho^{j-1} E_t(\Delta e_{t+j} - r_{t+j}) + (1-\rho)\sum_{j=1}^{\infty} \rho^{j-1} E_t(de_{t+j}). \tag{7}$$

This expression shows that the price-earnings multiple summarizes expectations about future earnings growth, future returns, and payout policy over many horizons. It therefore provides a natural object through which macroeconomic information can affect valuation, even when the direct relation between macroeconomic variables and next-period realized returns is weak. FA-SOP uses this implication in a parsimonious way: it does not estimate a full structural present-value model in the forecasting exercise, but uses macroeconomic factors to construct a conditional valuation level for pe_t and then forecasts gm_{t+1} through the correction of deviations

from that level.

2.3 Passive Sum-of-the-parts (SOP) restrictions

Given the decomposition above, the general SOP forecast can be written as

$$\widehat{r}_{t+1}^{SOP} = \widehat{gm}_{t+1} + \widehat{ge}_{t+1} + \widehat{dp}_{t+1}. \tag{8}$$

The core SOP specification, which we refer to as Simple SOP, is built on a set of passive restrictions. Expected earnings growth is estimated by a 20-year moving average of past earnings growth, denoted by $\overline{ge}_t^{20}$; the expected dividend-price component is set equal to the current dividend-price ratio, dp_t; and the expected price-earnings multiple growth forecast is restricted to zero, $\widehat{gm}_{t+1} = 0$.

Thus, the Simple SOP forecast[1] is

$$\widehat{r}_{t+1}^{Simple\text{-}SOP} = \overline{ge}_t^{20} + dp_t. \tag{9}$$

This simple rule is useful because it highlights the passive discipline embedded in the original SOP approach.[2] It exploits the return decomposition without estimating unstable return-prediction coefficients, thereby providing a low-estimation-error benchmark. Any more active component-level modeling must therefore deliver sufficient predictive gains to offset the additional estimation uncertainty it introduces.

FSC also consider a low-dimensional active variant that models gm_{t+1} through reversion in the price-earnings multiple. In this specification, the conditional valuation level for the log price-earnings multiple is estimated from a single observed predictor x_t:

$$pe_t = a + \omega x_t + u_t, \tag{10}$$

and the deviation $-\widehat{u}_t = \widehat{pe}_t - pe_t$ is used as the reversion signal in a second-stage regression,

$$gm_{t+1} = c + d(-\widehat{u}_t) + v_{t+1}. \tag{11}$$

[1] The corresponding equity premium forecast is obtained by subtracting the log risk-free rate from the Simple SOP return forecast: $\widehat{re}_{t+1}^{Simple\text{-}SOP} = \widehat{r}_{t+1}^{Simple\text{-}SOP} - r_{f,t+1}$, where $r_{f,t+1}$ is the log risk-free rate known at time t. The equity premium forecasts for all other SOP-type specifications are obtained in the same way.

[2] As noted in FSC, this formulation can also be interpreted as a restricted version of the traditional predictive regression: $r_{t+1} = \alpha + \beta\, dp_t + \varepsilon_{t+1}$ with $\alpha = \overline{ge}_t^{20}$ and $\beta = 1$.

We refer to this specification as SOP-reversion. It is the direct predecessor of FA-SOP: the key question is whether the conditional valuation level can be measured more accurately when the single predictor x_t is supplemented with latent macroeconomic factors.

2.4 Factor-Augmented-Sum-of-the-Parts (FA-SOP) method

FA-SOP extends the SOP-reversion logic by replacing the single-predictor estimate of the conditional valuation level with a factor-augmented estimate. The passive forecasts for earnings growth and the dividend-price component are retained, while active modeling is directed only to gm_{t+1}. This design turns the FSC tension into an empirical test: if active gm modeling failed because the conditional valuation level was measured too narrowly, then a broader macro-implied valuation level should improve forecasts; if gm contains little usable predictable variation, the factor-augmented step should add little.

The method proceeds in four steps.

Step 1. Extract latent macroeconomic factors. Let $\mathbf{X}_t$ denote a large macroeconomic dataset. We assume that its common variation can be summarized by a small number of latent factors:

$$X_{it} = \lambda_i' \mathbf{F}_t + \epsilon_{it}. \tag{12}$$

Here, $\mathbf{F}_t$ provides a parsimonious representation of the macroeconomic state embedded in the high-dimensional information set. The factors are estimated by principal components, as described in the estimation subsection below.

Step 2. Construct the valuation-adjustment signal. Let $pe_t = \log(P_t/E_t)$ denote the log price-earnings multiple and let x_t denote an observed economic state variable. We first estimate the macro-implied valuation level through the contemporaneous factor-augmented regression

$$pe_t = a + b'\mathbf{F}_t + \omega x_t + u_t. \tag{13}$$

The fitted value $\widehat{pe}_t = \widehat{a} + \widehat{b}'\widehat{\mathbf{F}}_t + \widehat{\omega} x_t$ represents the macro-implied valuation level for the price-earnings multiple. The valuation-adjustment signal is then defined as the deviation of this fitted level from the observed multiple:

$$-\widehat{u}_t = \widehat{pe}_t - pe_t. \tag{14}$$

A positive value of $-\widehat{u}_t$ means that the observed multiple is below its macro-implied valuation level, so reversion toward that level predicts positive subsequent multiple growth. A negative value indicates that the observed multiple is above its macro-implied level and therefore predicts negative subsequent multiple growth. The observable predictor x_t allows the fitted valuation level to incorporate theoretically motivated observed state variables. This specification therefore does not force all macro-financial information to enter only through the estimated latent factors.

Low-dimensional counterpart. When $\mathbf{F}_t$ is omitted, equation (13) reduces to the SOP-reversion first-stage specification in equation (10).

Step 3. Forecast price-earnings multiple growth. Because valuation adjustment may occur gradually, we link subsequent multiple growth to the valuation-adjustment signal through

$$gm_{t+1} = c + d(-\widehat{u}_t) + v_{t+1}. \tag{15}$$

We use the estimated coefficients (after applying Connor (1997) shrinkage as described in Online Appendix A) to forecast price-earnings multiple growth gm as

$$\widehat{gm}_{t+1}^{FA\text{-}Rev} = \widehat{c} + \widehat{d}(-\widehat{u}_t). \tag{16}$$

Step 4. Recompose the return forecast. Finally, FA-SOP preserves the passive SOP forecasts for the other two components and combines them with the factor-augmented forecast of multiple growth:

$$\widehat{r}_{t+1}^{FA\text{-}SOP} = \widehat{gm}_{t+1}^{FA\text{-}Rev} + \overline{ge}_t^{20} + dp_t. \tag{17}$$

This construction shows how FA-SOP implements the idea of "predicting the predictable" within the SOP decomposition: richer macroeconomic information is used only where the decomposition and present-value logic suggest the signal is most likely to appear.

2.5 Economic interpretation of FA-SOP

FA-SOP can be viewed as a targeted relaxation of the passive SOP restrictions. The forecasts for *dp* and *ge* remain passive: the dividend-price component is forecast by its current value, and earnings growth is forecast by a long moving average. These choices follow the low-estimation-error discipline of SOP and reflect the limited scope for stable short-horizon gains

in these two components.[3]

The active forecast is reserved for gm_{t+1}, the change in the price-earnings multiple, because this component captures valuation adjustment. FA-SOP uses the level of the multiple, pe_t, to construct the macro-implied valuation level in the first-stage regression. The deviation between this fitted level and the observed multiple,

$$-\widehat{u}_t = \widehat{pe}_t - pe_t,$$

is the valuation-adjustment signal used to forecast gm_{t+1}. When the observed multiple is below the level implied by macroeconomic conditions, subsequent multiple growth is expected to be higher; when it is above that level, subsequent multiple growth is expected to be lower. Thus, the first-stage level equation measures valuation misalignment, while the second-stage regression maps that misalignment into a forecast of valuation change.[4]

FSC's observable-predictor reversion specification is a low-dimensional version of this logic, but in their evidence this active step did not consistently improve on the passive SOP benchmark. FA-SOP extends the reversion approach by measuring the macro-implied valuation level with latent macroeconomic factors rather than with a single observed predictor.

2.6 Estimation

We describe the estimation procedure used in the first-stage regression of the FA-SOP method, which builds on the factor-augmented regression (FAR) framework. While FA-SOP adopts a contemporaneous specification, it also extends the standard FAR framework by allowing for a mixture of stationary and nonstationary factors in the forecasting equation. Specifically, we forecast pe_t using Hannadige et al. (2024)'s generalized factor structure, where the conventional FAR approach based solely on stationary factors—typically used for forecasting r_t—emerges as a special case. Since the latent factors are unobservable, estimation proceeds in two steps: first, principal components analysis is used to extract the factors; then, coefficients

[3] In our sample, the AR(1) coefficient of the dividend-price ratio is approximately 0.99 at the monthly frequency and 0.97 at the quarterly frequency. Annual earnings growth has an AR(1) coefficient close to zero. At the monthly frequency, earnings growth is based on overlapping 12-month earnings, which mechanically creates persistence across adjacent observations.

[4] This use of a level equation is also consistent with the time-series behavior of the price-earnings multiple. In our sample, the ADF test with an intercept does not reject the unit-root null for the price-earnings multiple: the p-values are 0.5928 at the monthly frequency and 0.4201 at the quarterly frequency. The number of lags is set to $\lfloor 12(T/100)^{1/4} \rfloor$, following Schwert (1989). Using fixed lags of 12 for monthly data and 4 for quarterly data gives p-values of 0.0759 and 0.0949, respectively, which also fail to reject the unit-root null at the 5% level. Thus, the forecast has an error-correction interpretation for changes in an (I(1)) valuation variable.

are estimated by ordinary least squares.

The number of nonstationary and stationary factors, p and q, is first determined using the Integrated Panel Criterion (IPC) proposed by Bai (2004), which selects the number of factors by minimizing the sum of squared residuals while applying a penalty term to ensure model parsimony. Let $\mathbf{X} = (\mathbf{X}_{it})_{T\times N}$ denote the panel data available at time T in matrix form, and let $D_T = \text{diag}\left(TI_p, T^{1/2}I_q\right)$. Denoting $k = p + q$, the estimator of $\mathbf{F}$, $\hat{\mathbf{F}}_{T\times k} = \left(\hat{F}_1, \ldots, \hat{F}_T\right)'$, is defined as D_T times the matrix formed by the k eigenvectors corresponding to the k largest eigenvalues of the matrix $\mathbf{XX}'$. We adopt the Augmented Dickey-Fuller (ADF) test to classify the columns of $\hat{\mathbf{F}}$ before they are multiplied by D_T, ensuring that the first p columns are identified as nonstationary (with their corresponding eigenvalues arranged in decreasing order), and the remaining q columns are identified as stationary, also ordered by decreasing eigenvalues. Once the factors have been estimated, the factor loading matrix $\mathbf{\Lambda} = (\lambda_1, \ldots, \lambda_N)'$ can be estimated by $\hat{\mathbf{\Lambda}} = \mathbf{X}'\hat{\mathbf{F}}D_T^{-2}$. It should be noted that we use the original dataset (which contains I(1) series) to estimate the unobservable factors in the FA-SOP method and use the transformed dataset (to ensure stationarity) to estimate unobservable factors in the factor-augmented regression that directly forecasts r_t.

After obtaining the estimator of the latent factors, $\hat{\mathbf{F}}_t$ based on $\mathbf{X}_t$, we obtain the coefficients by regressing the target variable, r_{t+1} or pe_t in equation (13), on a constant, $\hat{\mathbf{F}}_t$, and x_t.

2.7 Evaluation measures

2.7.1 Statistical gain

Following Campbell and Thompson (2008), we evaluate the out-of-sample forecasting performance using the out-of-sample R^2, denoted as R^2_{OOS},

$$R^2_{OOS} = 1 - \frac{\sum_{t=t_0}^{T}(r_t - \hat{r}_t)^2}{\sum_{t=t_0}^{T}(r_t - \bar{r}_t)^2}, \tag{18}$$

where r_t is the realized stock market return or equity premium at time t, $\hat{r}_t$ is the forecast from a predictive model estimated through period $t-1$, and $\bar{r}_t$ is the historical average return (or premium) up to time $t-1$. The comparison begins at time t_0 and continues through time T. The R^2_{OOS} statistic measures the proportional reduction in mean squared forecast error (MSFE) of the predictive model relative to the historical average benchmark. A positive R^2_{OOS} indicates that the predictive model outperforms the historical mean. The statistical significance of the

results is evaluated by the MSFE-adjusted statistic of Clark and West (2007), which tests the null of $H_0 : R^2_{OOS} \leq 0$ against $H_A : R^2_{OOS} > 0$.

2.7.2 Economic gain

To evaluate the economic value of different return forecasting methods, we follow Campbell and Thompson (2008), Ferreira and Santa-Clara (2011), and Neely et al. (2014), among others, by implementing out-of-sample trading strategies that allocate between equities and the risk-free asset. A mean-variance investor optimally allocates wealth at the end of time t based on the return forecast for period $t+1$. The share of the portfolio invested in equities is given by:

$$w_t = \frac{\hat{r}_{t+1} - r_{f,t+1}}{\gamma \hat{\sigma}^2_{t+1}}, \tag{19}$$

where $\hat{r}_{t+1}$ is the forecasted stock market return, $r_{f,t+1}$ is the log risk-free return from t to $t+1$ (known at time t), γ is the coefficient of relative risk aversion (set to 3), and $\hat{\sigma}^2_{t+1}$ is the forecasted variance of stock returns. The remaining fraction, $1 - w_t$, is allocated to risk-free assets. The portfolio return at time $t+1$ is then given by:

$$r_{p,t+1} = w_t r_{t+1} + r_{f,t+1}. \tag{20}$$

We assume that the investor uses a five-year moving window of past returns to estimate the variance of the equity risk premium and constrain w_t to lie between 0 and 1.5. Imposing such realistic bounds ensures more stable allocations, thereby mitigating the well-known sensitivity of mean-variance optimal weights to return forecasts. This setup follows common practice in the return predictability literature and facilitates comparison with prior studies (Rapach et al., 2010; Neely et al., 2014; Rapach et al., 2016; Cheng et al., 2025).[5]

We use two measures to assess the performance of the asset allocation strategy. The first criterion, certainty equivalent return (CER) is defined as

$$CER_p = \hat{\mu}_p - \frac{\gamma}{2}\hat{\sigma}^2_p, \tag{21}$$

where $\hat{\mu}_p$ and $\hat{\sigma}^2_p$ are the mean and variance for the investor's portfolio return over the forecast

[5]FSC set $\gamma = 2$, estimate $\hat{\sigma}_{t+1}$ using all data available up to time t, and do not impose bounds on w_t. We adopt a more conservative specification to ensure that the reported economic gains are not driven by extreme portfolio allocations. In Online Appendix C, we further show that our main results remain robust under the original FSC criteria.

evaluation period, respectively. The CER can be interpreted as the risk-free rate of return that an investor is willing to accept using a certain predictive model, instead of adopting the given risky portfolio. The second criterion is the Sharpe ratio (SR),

$$SR_p = \frac{\hat{\mu}_p - \bar{r}_f}{\bar{\sigma}_p} \tag{22}$$

where $\hat{\mu}_p - \bar{r}_f$ and $\bar{\sigma}_p$ are the mean and standard deviation of portfolio excess returns, respectively. The Sharpe ratio measures how much excess return the portfolio generates for each unit of total risk the investor takes.

We report the CER gain and Sharpe ratio gain, which represent the portfolio performance improvements from using a predictive model relative to the historical mean (HM) benchmark. The CER gain reflects the fee an investor would be willing to pay to use the model's forecasts, while the Sharpe ratio gain captures the increase in return per unit of risk. All results are annualized to facilitate consistent comparison.

3 Data

Stock returns are calculated from the S&P 500 index and are inclusive of dividends. The risk-free rate is proxied by the Treasury bill rate, which is used to compute the equity premium (presented in Online Appendix E). Both stock returns and T-bill rates are expressed in continuously compounded terms. The data for these series are obtained from Amit Goyal's website.[6] We consider two data frequencies—monthly and quarterly—over the sample period from January 1960 to December 2022, yielding 756 monthly and 252 quarterly observations.

Table 1 reports summary statistics for the stock market return (r) and its components (gm, ge, and dp), along with their pairwise correlations at both monthly and quarterly frequencies. The mean return is 0.79% per month and 2.37% per quarter, and the distribution is left-skewed with excess kurtosis. Among the components, ge and dp have significantly higher means than gm, while dp exhibits negligible volatility compared to the much larger standard deviations of gm and ge. Panel B of Table 1 shows that stock returns are strongly negatively correlated with ge and moderately positively correlated with gm, while the other correlations are relatively weak.

We obtain 14 widely studied economic variables from the updated dataset of Welch and

[6]We thank Professor Goyal for making the data available at https://sites.google.com/view/agoyal145.

Table 1: Summary statistics of return components

Panel A: Univariate statistics								
Return components	Mean	Median	Standard deviation	Minimum	Maximum	Skewness	Kurtosis	AR(1)
Monthly frequency (January 1960–December 2022)								
gm	0.03	0.10	6.24	−72.38	53.61	−1.22	37.75	0.35
ge	0.52	0.68	4.92	−52.83	70.39	1.87	88.77	0.75
dp	0.24	0.24	0.09	0.09	0.53	0.63	2.71	0.99
r	0.79	1.20	4.33	−24.25	15.51	−0.65	5.24	0.03
Quarterly frequency (1st quarter 1960–4th quarter 2022)								
gm	0.10	0.36	23.10	−141.62	112.12	−0.55	12.22	−0.37
ge	1.55	2.43	22.81	−109.55	137.96	0.02	13.27	−0.33
dp	0.72	0.72	0.28	0.27	1.54	0.66	2.82	0.97
r	2.37	3.41	8.14	−28.87	20.64	−0.94	4.68	0.05

Panel B: Correlations				
Return components	*gm*	*ge*	*dp*	*r*
Monthly frequency (January 1960–December 2022)				
gm	1			
ge	−0.72	1		
dp	0.01	−0.07	1	
r	0.62	0.1	−0.04	1
Quarterly frequency (1st quarter 1960–4th quarter 2022)				
gm	1			
ge	−0.94	1		
dp	0.01	−0.05	1	
r	0.21	0.14	−0.09	1

Note: Panel A reports summary statistics for the realized components of stock market returns, including the mean, median, standard deviation, minimum, maximum, skewness, kurtosis, and the first-order autocorrelation coefficient (AR(1)). Specifically, *gm* denotes growth in the price-earnings multiple, *ge* denotes earnings growth, *dp* is the dividend-price ratio, and *r* represents the overall stock market return. Panel B presents the correlation matrix of these return components. The sample period spans from January 1960 to December 2022.

Goyal (2008), which have been frequently used in the stock return prediction literature (Rapach et al., 2010; Ferreira and Santa-Clara, 2011; Dangl and Halling, 2012; Neely et al., 2014). These variables serve as observable predictors in the SOP, FA-Unified, and FA-SOP methods. Predictor definitions are reported in Online Appendix I.

We employ two large-scale macroeconomic datasets to estimate the monthly and quarterly macroeconomic factors. For the monthly analysis, we use 116 variables from the FRED-MD database, as described in McCracken and Ng (2016). For the quarterly analysis, we use the FRED-QD dataset introduced in McCracken and Ng (2021), which contains 227 variables. Detailed descriptions of both datasets can be found in Online Appendix I or on Professor McCracken's website[7]. One might argue that macroeconomic data are subject to future revisions and not fully available in real time, potentially biasing our interpretation of predictability. However, in the context of a factor structure, this concern is largely mitigated. Since the factors are extracted from a large number of macroeconomic series—many of which have independent or weakly correlated revisions—the idiosyncratic noise introduced by revisions tends to average out. As a result, the common factors remain largely unaffected. Bernanke and Boivin (2003)

[7] https://research.stlouisfed.org/econ/mccracken/fred-databases/

provide empirical support for this view, showing that forecasts based on factor models are not sensitive to whether revised or real-time data are used.

Before extracting factors, we preprocess the macroeconomic panels to handle missing observations, outliers, and stationarity transformations. Missing observations are imputed using the expectation-maximization procedure of Stock and Watson (2002) as implemented by McCracken and Ng (2016). For FA-Unified, which forecasts stationary returns directly, we use the stationarity-transformed macroeconomic panels following the variable-specific transformation codes in McCracken and Ng (2016, 2021). For FA-SOP, which models the persistent price-earnings multiple, we extract factors from the raw panels and allow both stationary and nonstationary factors to enter the fitted valuation level. Additional preprocessing details and variable lists are provided in Online Appendix I.

4 Results

This section evaluates the central target-design prediction: the same underlying macroeconomic panel should be more useful when directed toward valuation adjustment than when used to forecast aggregate returns directly. We first describe the head-to-head comparison design that separates the value of decomposition from the value of the information set. We then examine the extracted latent factors and compare the statistical return predictability of FA-SOP with that of standard benchmarks, including the historical mean, univariate predictive regressions, FA-Unified, Simple SOP, and SOP-reversion. We finally assess the economic value of these forecasts through portfolio allocation exercises.

The main analysis is based on log return forecasts, consistent with the return decomposition in Section 2. Results for simple returns and equity premia are qualitatively similar, and in some cases stronger, and are reported in Online Appendices D and E.

4.1 Empirical design: head-to-head comparisons

A central empirical question is whether the relative performance of FA-SOP comes from the decomposition structure itself or simply from using a richer macroeconomic information set. The key benchmark for this question is FA-Unified, the direct-return counterpart to FA-SOP. It holds the broad macroeconomic panel fixed while changing the forecasting target.

FA-Unified applies the information directly to aggregate returns, whereas FA-SOP first

organizes returns through the SOP decomposition and then applies the information to price-earnings multiple growth, gm_{t+1}. The comparison therefore asks whether decomposition changes the usefulness of the same information, rather than whether a richer information set mechanically improves forecasts.

For completeness, we also compare FA-SOP with the historical mean, univariate predictive regressions, Simple SOP, and the observable-predictor SOP-reversion benchmark. The historical mean and univariate predictive regressions are standard benchmarks in the return-predictability literature. Simple SOP captures the passive low-estimation-error benchmark of Ferreira and Santa-Clara (2011), while SOP-reversion provides the low-dimensional observable-predictor counterpart to FA-SOP's factor-augmented valuation adjustment.

4.2 Latent factors

Before turning to forecast performance, we briefly summarize the factor estimates used in the empirical analysis. The IPC_2 criterion of Bai (2004) selects eight factors for both the monthly and quarterly datasets. For FA-SOP, the ADF classification described in Section 2 identifies four nonstationary and four stationary factors; FA-Unified uses stationary factors extracted from the transformed macroeconomic panels. Online Appendix C plots the estimated monthly factors used in both implementations.

4.3 Return predictability

We begin with a short in-sample diagnostic for the traditional univariate regression and the FA-Unified regression.[8] The univariate predictive regression estimates $r_{t+1} = \alpha + \beta x_t + \epsilon_{t+1}$, where x_t is a single lagged observable variable. The FA-Unified regression adds latent macroeconomic factors to the same type of direct return-forecasting equation. The second and third columns of Table 2 report the corresponding full-sample adjusted R^2 values.

The diagnostic confirms that macroeconomic factors contain in-sample explanatory power for returns, consistent with Ludvigson and Ng (2007): FA-Unified has substantially higher adjusted R^2 values than the traditional univariate regressions. This evidence is descriptive rather than central to our evaluation. It shows that the latent factors are not devoid of return-related information. The out-of-sample exercise below asks whether this information can be

[8]The SOP and FA-SOP methods are designed for out-of-sample return prediction. We report the in-sample performance of the univariate predictive regression and FA-Unified regression only to document whether macroeconomic factors have explanatory content before turning to the out-of-sample forecasting exercise.

used reliably, and whether its usefulness depends on the forecasting structure.

We evaluate the out-of-sample forecasting performance of the FA-SOP method alongside five benchmark approaches. The out-of-sample period begins 20 years after the start of the sample, and all methods—except for the Simple SOP—are estimated using an expanding window. The Simple SOP does not involve parameter estimation, so it does not require a recursive window.[9] The fourth to ninth columns of Table 2 report the R^2_{OOS} values across methods and predictor choices. For FA-SOP, each row corresponds to the observable variable x_t included in the first-stage valuation regression, with the row labeled "Constant" denoting the specification that uses only latent macroeconomic factors.

We summarize the results for the two versions of the univariate predictive regression as follows. First, consistent with Welch and Goyal (2008), the univariate predictive regression yields negative R^2_{OOS} values in most cases, ranging from -2.28% to -0.17% at the monthly frequency and from -22.54% to -0.76% at the quarterly frequency. The only exception is an insignificant positive R^2 of 0.31% when using the long-term government bond return (LTR) as the predictor at the quarterly frequency. Second, applying shrinkage (Connor, 1997) improves the out-of-sample performance for all predictors. As shown in the fifth column of Table 2, five out of fourteen variables produce positive R^2_{OOS} values at the monthly frequency, and six do so at the quarterly frequency. However, the improvements remain modest and statistically insignificant, with maximum values of 0.26% and 0.33% at the monthly and quarterly frequencies, respectively.

We now turn to the out-of-sample prediction results of the FA-Unified, SOP, and FA-SOP methods. As shown in the sixth column of Table 2, the FA-Unified regression consistently produces negative out-of-sample R^2 values across all specifications, regardless of the additional economic variable used. The underperformance is especially pronounced at the quarterly frequency, where R^2_{OOS} values range from -42.03% to -12.00%.

The seventh and eighth columns of Table 2 present the results for the simple SOP and SOP with multiple reversion methods. The simple SOP, which assumes no multiple growth, yields positive out-of-sample R^2 values of 0.32% at the monthly frequency and 2.17% at the quarterly frequency, though the monthly result is not statistically significant.[10] Using alternative variables

[9]The univariate predictive regression follows Welch and Goyal (2008), while the shrinkage version adopts the approach of Connor (1997), imposing a prior of no predictability by shrinking the slope coefficient toward zero and setting the intercept to the unconditional mean of returns. This prior is also used in the second-stage regression of both the SOP-reversion and FA-SOP methods.

[10]For the SOP with multiple reversion and FA-SOP, we follow Ferreira and Santa-Clara (2011) and exclude predictors that directly depend on stock prices—such as earnings-price ratio (EP), dividend-price ratio (DP), dividend yield (DY), and book-to-market ratio (BM)—to avoid explaining the price-earnings multiple with other

Table 2: Stock return prediction results

	In-sample R^2		R^2_{OOS}					
Variable	Univariate regression	FA-Unified regression	Univariate regression	Univariate regression (Shrinkage)	FA-Unified regression	SOP no multiple growth	SOP multiple reversion	FA-SOP multiple reversion
Panel A: Monthly return forecasts (January 1980–December 2022)								
SVAR	0.05	2.39***	−2.28	−0.39	−3.30	–	0.61*	1.94***
DFR	−0.02	1.78***	−0.77	−0.18	−3.10	–	0.64*	1.90***
LTY	−0.13	1.66***	−0.50	−0.10	−2.99	–	0.79**	1.77***
LTR	0.62**	1.59***	−0.32	0.26	−3.65	–	0.63*	1.95***
INFL	0.06	1.71***	−0.32	−0.03	−2.81	–	0.59*	1.66***
TMS	0.00	1.81***	−0.48	−0.02	−2.50	–	0.55*	1.94***
TBL	−0.13	1.90***	−0.72	−0.13	−2.92	–	0.79**	1.92***
DFY	0.16	1.61***	−0.73	−0.07	−3.23	–	0.41*	1.95***
NTIS	−0.11	1.55***	−0.59	0.01	−3.10	–	0.47*	1.88***
DE	−0.08	1.59***	−0.84	−0.19	−3.57	–	−0.23	0.45*
EP	0.13	2.63***	−0.17	0.06	−1.88	–	–	–
DP	0.39**	2.40***	−0.28	0.13	−2.51	–	–	–
DY	0.43**	2.41***	−0.31	0.14	−2.51	–	–	–
BM	0.07	2.14***	−0.29	−0.01	−2.71	–	–	–
Constant	–	1.68***	–	–	−2.47	0.32	0.63**	1.95***
Panel B: Quarterly return forecasts (1st quarter 1980–4th quarter 2022)								
SVAR	−0.07	2.25***	−22.54	−0.18	−42.03	–	2.87**	3.91**
DFR	−0.27	0.72***	−3.28	−0.15	−15.11	–	2.14*	5.79***
LTY	−0.35	0.80***	−1.35	−0.11	−17.23	–	3.31**	4.77**
LTR	1.64**	2.79***	0.31	0.33	−12.00	–	2.35*	5.58***
INFL	−0.18	0.74***	−0.76	−0.06	−17.37	–	3.30**	6.09***
TMS	−0.21	1.11***	−1.79	−0.07	−13.25	–	2.15*	4.54**
TBL	−0.40	1.10***	−2.13	−0.15	−15.19	–	3.02**	4.64**
DFY	0.37	0.99***	−2.26	−0.08	−16.26	–	2.40*	4.58**
NTIS	−0.33	0.74***	−1.78	0.00	−14.51	–	2.21*	4.46**
DE	−0.03	0.71***	−2.11	−0.18	−16.10	–	0.72	−2.89
EP	0.21	4.17***	−1.16	0.05	−15.01	–	–	–
DP	1.32**	4.16***	−1.16	0.26	−17.12	–	–	–
DY	1.57**	3.48***	−1.64	0.31	−17.11	–	–	–
BM	0.28	2.30***	−0.96	0.02	−18.10	–	–	–
Constant	–	1.10***	–	–	−12.88	2.17**	2.24*	4.88**

Note: This table presents the in-sample (adjusted) and R^2_{OOS} values (in percentage) for stock market return forecasts at the monthly and quarterly frequencies from various methods. The in-sample R^2 is estimated over the full-sample period. The R^2_{OOS} compares the forecast error of the model with the forecast error of the historical mean. The sample period is from January 1960 through December 2022. Forecasts begin 20 years after the sample start. Following Ferreira and Santa-Clara (2011), the predictors that directly depend on the stock index price (earnings price, EP; dividend price, DP; dividend yield, DY; and book-to-market, BM) are not used. Asterisks denote significance of the R^2_{OOS} as measured by the CW-test statistic of Clark and West (2007). ***, ** and * denote significance at the 1%, 5% and 10% level, respectively.

to capture the state of the economy, the SOP with multiple reversion method achieves positive out-of-sample R^2 values ranging from 0.41% to 0.79% at the monthly frequency and from 2.14% to 3.31% at the quarterly frequency (excluding the dividend payout ratio, DE). All results mentioned above, except for DE, are statistically significant at the 10% level, with several also significant at the 5% level.[11]

multiples.

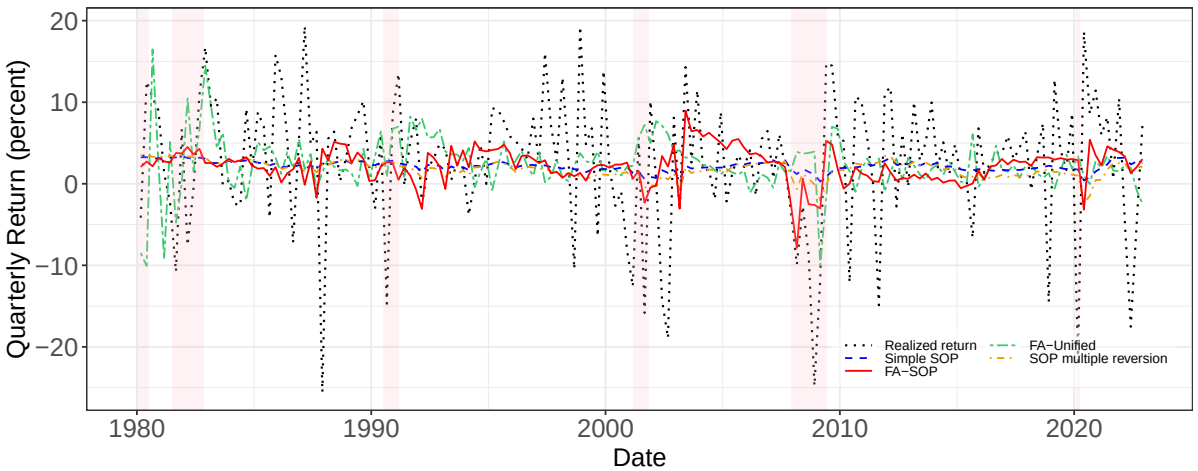
[11] The SOP results are not directly comparable to Ferreira and Santa-Clara (2011) because our sample starts in January 1960, constrained by FRED-MD/FRED-QD availability, and extends through December 2022, whereas their sample runs from December 1927 to December 2007. The lower SOP performance in our sample mainly reflects the different evaluation period, especially the inclusion of the post-2007 period in which the passive SOP rule delivers smaller gains.

The last column of Table 2 reports the results of FA-SOP. Even when using only macroeconomic factors to capture the state of the economy (the "Constant" row), the method achieves an out-of-sample R^2 of 1.95% at the monthly frequency—nearly three times higher than that of the SOP with multiple reversion. At the quarterly frequency, FA-SOP also outperforms all other methods, delivering R^2 values ranging from 3.91% to 6.09%, except when the dividend payout ratio (DE) is used as the additional predictor. All FA-SOP results (excluding DE) are statistically significant at the 5% level, with most significant at the 1% level.

Figure 2 provides a visual comparison of return forecasts from the FA-SOP method against those from the SOP and FA-Unified methods. For simplicity, we plot "FA-Unified (Constant)", "SOP multiple reversion (Constant)", and "FA-SOP multiple reversion (Constant)". Since monthly forecast variation is relatively small compared to realized returns, we present quarterly results to enhance visual clarity.

The forecasts differ markedly across methods. FA-Unified forecasts fluctuate significantly before 2000 and closely track realized returns during the early sample period (e.g., 1980-1985), but their accuracy deteriorates in later years. In contrast, forecasts from the simple SOP and SOP with multiple reversion show limited variation throughout the sample, raising doubts about their responsiveness to return dynamics. FA-SOP forecasts display moderate variation and better alignment with realized returns in several periods, such as 1985-1990, 2007-2010, and post-2020.

Figure 2: Quarterly return forecast



Note: This figure shows the quarterly realized return and the forecast of market return from the FA-Unified, simple SOP, SOP with multiple reversion and FA-SOP methods. The vertical shaded area represents the business cycle recession dates reported by NBER.

In summary, the univariate predictive regression performs poorly out-of-sample regardless

of the choice of observable predictor—most R^2_{OOS} values are negative and statistically insignificant, consistent with the findings of Welch and Goyal (2008) and Goyal et al. (2024). FA-Unified performs even worse, despite using the same underlying macroeconomic panel as FA-SOP. In contrast, FA-SOP delivers strong out-of-sample performance, achieving a maximum R^2_{OOS} of 1.95% at the monthly frequency and up to 6.09% at the quarterly frequency. These gains are nearly double those of the best-performing SOP specification. The evidence therefore points to target organization, not the raw availability of macroeconomic information, as the key difference across models.

4.4 Economic predictability

We further evaluate the economic value of return forecasts by computing the annualized certainty-equivalent return (CER) gains and Sharpe ratio gains for a mean-variance investor with a risk aversion coefficient of $\gamma = 3$, who allocates optimally between equities and the risk-free asset using out-of-sample predictions. Tables 3 and 4 report the results for the FA-Unified, SOP, and FA-SOP methods, both with and without a proportional transaction cost of 50 basis points per trade.

FA-SOP consistently delivers the highest CER gains across the vast majority of settings, with especially strong performance at the monthly horizon. At the monthly frequency, FA-Unified generates pre-cost CER gains between -0.28% and 3.13%, but most of these gains become negative once trading costs are included. SOP methods perform modestly better, capping at a post-cost CER gain of 1.30%. FA-SOP, by comparison, achieves CER gains exceeding 3.00% in nearly all cases without transaction costs (except when using the dividend payout ratio, DE), and retains strong performance even after accounting for 50 bps transaction costs—often delivering gains more than double those of competing methods.

The Sharpe ratio gains closely mirror the patterns observed in the CER analysis. At the monthly frequency, the FA-Unified again demonstrates limited economic value, turning negative post-costs. Conversely, FA-SOP delivers substantial Sharpe ratio gains ranging from 0.20 to 0.26 after transaction costs, consistently doubling the performance of the univariate SOP-reversion methods.

At the quarterly frequency, FA-Unified continues to perform poorly, with uniformly negative economic gains, whereas both SOP and FA-SOP generate positive CER and Sharpe-ratio improvements. FA-SOP attains the highest pre-cost CER gain, reaching 3.99% when augmented

Table 3: Certainty Equivalent Return gains of the return forecasts.

	No trans. cost				50 bps trans. cost			
Variable	FA-Unified regression	SOP no multiple growth	SOP multiple reversion	FA-SOP multiple reversion	FA-Unified regression	SOP no multiple growth	SOP multiple reversion	FA-SOP multiple reversion
Panel A: Monthly return forecasts (January 1980–December 2022)								
SVAR	3.13	–	0.93	3.57	0.16	–	0.67	3.11
DFR	1.93	–	1.51	3.64	−1.11	–	1.30	3.17
LTY	0.70	–	1.45	3.49	−2.49	–	1.28	3.03
LTR	1.10	–	1.48	3.73	−1.88	–	1.27	3.26
INFL	1.48	–	1.97	3.63	−1.52	–	1.22	3.18
TMS	0.82	–	1.25	3.53	−2.00	–	1.04	3.05
TBL	0.93	–	1.31	3.86	−2.34	–	1.12	3.38
DFY	0.92	–	0.46	3.62	−2.20	–	0.30	3.17
NTIS	1.34	–	1.27	3.40	−1.45	–	1.08	2.92
DE	1.95	–	0.47	1.22	−0.46	–	0.31	0.59
EP	2.35	–	–	–	−0.55	–	–	–
DP	−0.24	–	–	–	−2.88	–	–	–
DY	−0.13	–	–	–	−2.88	–	–	–
BM	−0.28	–	–	–	−3.27	–	–	–
Constant	1.46	1.10	1.49	3.62	−1.44	0.96	1.28	3.17
Panel B: Quarterly return forecasts (1st quarter 1980–4th quarter 2022)								
SVAR	−2.53	–	3.43	2.45	−3.47	–	3.16	1.94
DFR	−2.06	–	3.10	2.84	−3.06	–	2.83	2.31
LTY	−2.57	–	3.19	2.78	−3.64	–	2.93	2.28
LTR	−1.70	–	3.29	2.87	−2.65	–	3.01	2.34
INFL	−2.70	–	3.90	3.99	−3.64	–	3.51	3.38
TMS	−1.92	–	3.20	2.55	−2.82	–	2.92	2.03
TBL	−2.70	–	3.25	2.63	−3.69	–	2.94	2.11
DFY	−2.46	–	3.32	2.68	−3.44	–	3.07	2.19
NTIS	−1.63	–	3.16	2.61	−2.44	–	2.91	2.10
DE	−1.51	–	1.93	0.26	−2.38	–	1.62	−0.34
EP	−1.91	–	–	–	−2.84	–	–	–
DP	−1.91	–	–	–	−2.63	–	–	–
DY	−2.25	–	–	–	−2.93	–	–	–
BM	−2.36	–	–	–	−3.15	–	–	–
Constant	−1.82	2.20	3.17	2.75	−2.76	1.97	2.90	2.24

Note: This table reports the annualized Certainty Equivalent Return (CER) gains, in percentage terms, from forecasting stock market returns at monthly and quarterly frequencies using various methods. A mean-variance investor with a risk-aversion coefficient of $\gamma = 3$ is assumed. Results are shown both with and without a 50 basis point (bps) proportional transaction cost. The sample period spans January 1960 to December 2022, with forecasts beginning 20 years after the sample start. Following Ferreira and Santa-Clara (2011), predictors that are functions of the stock price—namely earnings-price (EP), dividend-price (DP), dividend yield (DY), and book-to-market (BM)—are excluded to avoid mechanical dependencies.

with inflation (INFL). Averaged across predictors, however, FA-SOP performs comparably to, rather than uniformly better than, the univariate SOP specifications in the quarterly allocation exercise. This contrast highlights that forecast accuracy and portfolio performance are linked through the allocation rule, rather than mechanically equivalent. By extracting dense information from high-dimensional macroeconomic datasets, FA-SOP produces more responsive expected-return signals. These signals improve statistical tracking, but they may also translate into larger changes in portfolio weights and more frequent movements toward the imposed bounds. At the quarterly frequency, such positions are held fixed for three months and may therefore be more exposed to interim return reversals and, when transaction costs are imposed,

Table 4: Sharpe ratio gains of the return forecasts

	No trans. cost				50 bps trans. cost			
Variable	FA-Unified regression	SOP no multiple growth	SOP multiple reversion	FA-SOP multiple reversion	FA-Unified regression	SOP no multiple growth	SOP multiple reversion	FA-SOP multiple reversion
Panel A: Monthly return forecasts (January 1980–December 2022)								
SVAR	0.20	–	0.07	0.25	0.01	–	0.04	0.21
DFR	0.12	–	0.12	0.26	−0.07	–	0.10	0.22
LTY	0.04	–	0.10	0.24	−0.15	–	0.08	0.21
LTR	0.07	–	0.12	0.26	−0.10	–	0.09	0.23
INFL	0.09	–	0.18	0.26	−0.09	–	0.10	0.22
TMS	0.05	–	0.10	0.25	−0.11	–	0.07	0.21
TBL	0.06	–	0.09	0.28	−0.16	–	0.07	0.24
DFY	0.06	–	0.02	0.26	−0.13	–	0.01	0.22
NTIS	0.09	–	0.09	0.24	−0.07	–	0.07	0.20
DE	0.12	–	0.04	0.08	−0.02	–	0.01	0.03
EP	0.16	–	–	–	−0.05	–	–	–
DP	−0.02	–	–	–	−0.21	–	–	–
DY	−0.02	–	–	–	−0.21	–	–	–
BM	−0.02	–	–	–	−0.23	–	–	–
Constant	0.09	0.07	0.12	0.26	−0.08	0.06	0.09	0.22
Panel B: Quarterly return forecasts (1st quarter 1980–4th quarter 2022)								
SVAR	−0.12	–	0.29	0.14	−0.17	–	0.26	0.11
DFR	−0.11	–	0.27	0.17	−0.17	–	0.23	0.13
LTY	−0.15	–	0.21	0.16	−0.22	–	0.19	0.13
LTR	−0.08	–	0.29	0.17	−0.14	–	0.25	0.13
INFL	−0.15	–	0.31	0.24	−0.21	–	0.27	0.20
TMS	−0.09	–	0.27	0.15	−0.14	–	0.24	0.11
TBL	−0.16	–	0.21	0.15	−0.22	–	0.18	0.12
DFY	−0.14	–	0.29	0.16	−0.20	–	0.26	0.12
NTIS	−0.07	–	0.27	0.15	−0.11	–	0.24	0.12
DE	−0.06	–	0.14	0.00	−0.10	–	0.11	−0.03
EP	−0.13	–	–	–	−0.20	–	–	–
DP	−0.13	–	–	–	−0.18	–	–	–
DY	−0.16	–	–	–	−0.20	–	–	–
BM	−0.15	–	–	–	−0.20	–	–	–
Constant	−0.10	0.14	0.27	0.16	−0.15	0.12	0.24	0.13

Note: This table reports the Sharpe ratio gains from stock market return forecasts at monthly and quarterly frequencies using various methods. The investor is assumed to have a constant relative risk aversion of $\gamma = 3$, and all Sharpe ratio gains are annualized. Results are shown both without transaction costs and with a 50 basis points (bps) transaction cost per trade. The sample period spans from January 1960 to December 2022, with forecasts beginning 20 years after the sample start. Following Ferreira and Santa-Clara (2011), predictors directly tied to the stock index price—namely, the earnings-price ratio (EP), dividend-price ratio (DP), dividend yield (DY), and book-to-market ratio (BM)—are excluded.

larger rebalancing costs.

By contrast, the univariate SOP reversion model uses only one observed predictor and therefore produces smoother forecasts with lower time-series variation. In portfolio terms, these smoother signals can operate as a conservative regularization device, limiting large changes in portfolio weights and reducing realized portfolio volatility at lower rebalancing frequencies. Thus, the competitive quarterly Sharpe ratios of the simpler SOP specifications do not overturn the statistical advantage of FA-SOP. Rather, they indicate that the economic value of richer information depends on how forecast variation is translated into implementable portfolio positions. The machine learning evidence in Section 5.3 is consistent with this interpretation:

SOP-Ridge delivers strong quarterly CER and Sharpe-ratio gains by regularizing the contribution of the rich macroeconomic information set within the decomposition framework, while SOP-PLS and SOP-SVM show similar improvements.

In summary, this subsection shows that FA-SOP delivers substantial economic gains over the sample period and generally improves portfolio performance relative to the benchmark models. These findings are broadly consistent with the statistical evidence, while the quarterly results also highlight that forecast improvements must be translated into portfolio gains through an implementable allocation rule. The economic evidence therefore supports the target-design interpretation, but also makes clear that statistical improvements and portfolio gains need not be identical objects.

5 Sources of predictability

The results above establish the main empirical comparison: macroeconomic information is weak in direct aggregate-return forecasts but useful inside the decomposition. This section asks why. We organize the evidence around three linked questions. First, is valuation adjustment the margin through which predictability is recovered? Second, does the measurement of the macro-implied valuation level require latent factors rather than a small set of observable proxies? Third, does the same target-design principle survive when the forecasting model is made nonlinear and more flexible?

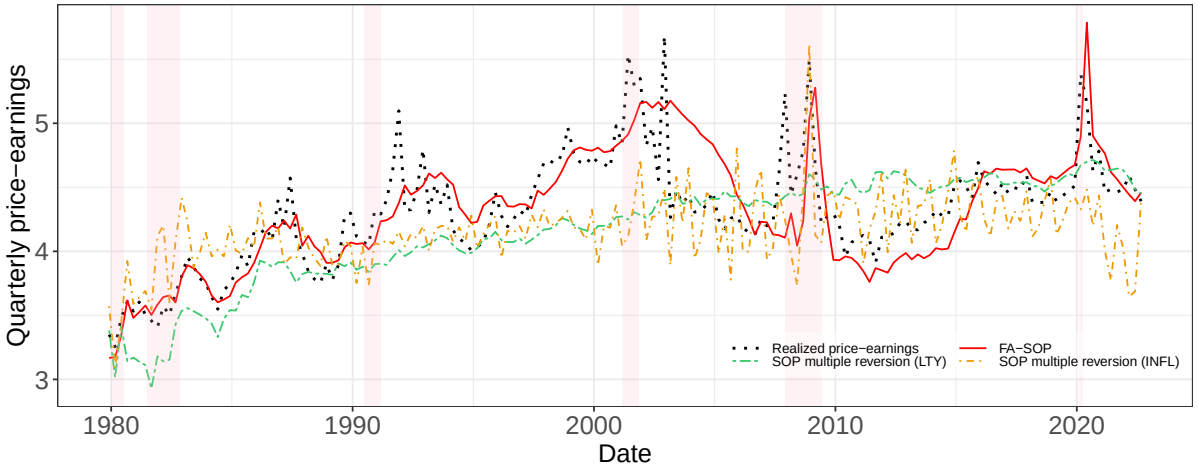
5.1 "Predicting the predictable" through valuation multiple reversion

The first mechanism is valuation multiple reversion. Although the object we ultimately forecast is the growth in the multiple, gm_{t+1}, the relevant economic mechanism operates through the level of the multiple, pe_t, and its reversion toward a conditional expectation determined by the macroeconomic state. Unlike the dividend-price ratio or earnings growth, which are either highly persistent or difficult to forecast at short horizons, the price-earnings multiple is a forward-looking valuation object that absorbs changes in discount rates, expected growth, and investor risk appetite. For this reason, it provides the natural margin along which macroeconomic information can become return predictability.

As discussed earlier, both the SOP with multiple reversion and the FA-SOP method assume that the price-earnings multiple reverts toward a conditional expectation. For return forecasting,

however, the relevant issue is not simply whether such reversion exists, but whether the deviation from the conditional expectation is corrected fast enough to generate useful forecasts of gm_{t+1}. A very slow reversion process may be consistent with valuation adjustment in the long run, but it provides little exploitable signal for out-of-sample return prediction. Figure 3 illustrates this point by plotting the quarterly realized price-earnings multiple together with fitted values from the first-stage regressions used in the SOP and FA-SOP methods. Specifically, we compare three specifications: two SOP-based first-stage regressions using a single predictor—the long-term government bond yield (LTY) and inflation (INFL)—and one FA-SOP first-stage regression using macroeconomic factors as predictors.

Figure 3: Quarterly price-earnings multiple forecast



Note: This figure shows the quarterly realized and fitted price-earnings multiple. It presents the fitted values from each estimation period in the first step of the multiple-reversion approach, $pe_t = a + \omega x_t + u_t$ for SOP-reversion and $pe_t = a + b'\mathbf{F}_t + \omega x_t + u_t$ for FA-SOP, which are used to calculate the residuals for estimating $\widehat{gm}_{t+1}$ in the second-stage regression (15). The vertical shaded area represents the business cycle recession dates reported by NBER.

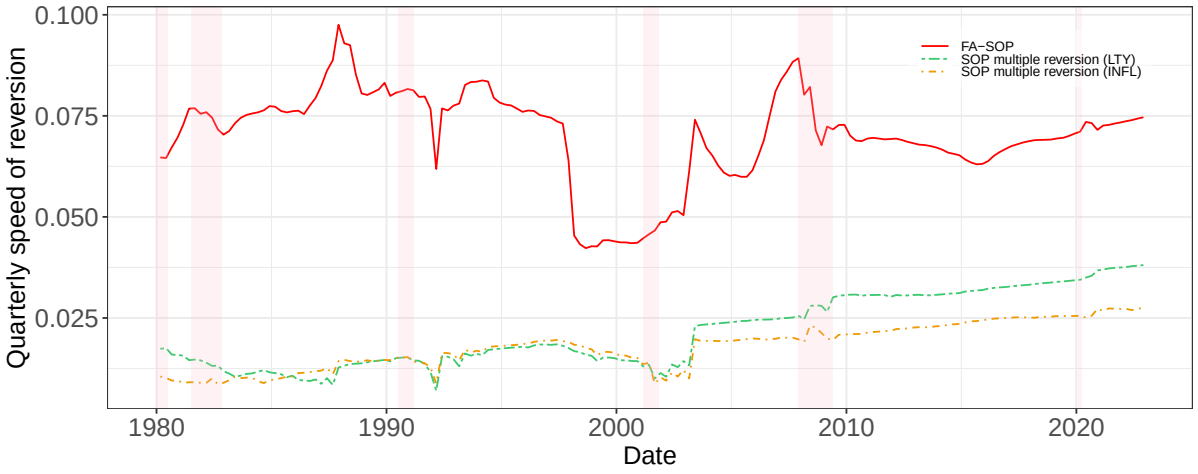
The figure reveals clear differences in how the fitted multiples behave across the three models. The SOP-based fitted lines show some alignment with the realized series during 1985–1995, suggesting partial reversion to expectations formed in earlier periods. However, this relationship weakens noticeably after 1995, with little evidence of sustained reversion. In contrast, the FA-SOP fitted multiple tracks the realized series more closely and crosses the realized series more frequently. These crossings are important because they indicate that valuation gaps are not merely persistent deviations from a fitted level; instead, they are followed by subsequent corrections. The relatively short-lived deviations around the FA-SOP fitted line therefore point to a faster adjustment process, rather than simply to a better static fit.

This distinction is especially important because the price-earnings multiple exhibits non-

stationary behavior in finite samples, so reversion is not mechanically guaranteed. In this setting, the first-stage FA-SOP specification does more than generate fitted values for pe_t: it uses nonstationary latent macroeconomic factors to construct a conditional valuation level. The estimated deviation, $-\hat{u}_t = \widehat{pe}_t - pe_t$, can then be used as an error-correction signal. The predictive content of gm_{t+1} arises from the subsequent closing of this deviation: when the observed multiple moves away from its conditional expectation, future multiple growth tends to move in the direction that restores the long-run relation. Since the fitted lines in Figure 3 are generated in a real-time, out-of-sample fashion using only information available up to each point in time, the figure provides evidence that this reversion signal is more timely under FA-SOP than under the single-predictor SOP specifications.

The importance of this mechanism is further illustrated by Figure 4, which reports the time-varying reversion speed captured by the coefficient d in equation (15). This coefficient governs how rapidly valuation deviations are corrected once the multiple moves away from its conditional expectation. On average, the reversion speed under FA-SOP is approximately three times that of the traditional SOP method. Economically, this implies that pricing errors are far less persistent under FA-SOP: when macroeconomic news or shifts in sentiment drive valuations away from their long-run relation, the subsequent adjustment is substantially faster.

Figure 4: Time-varying speed of price-earnings multiple reversion across methods



Note: This figure presents the coefficient d of the second stage regression (equation (15)) that characterizes the quarterly speed of reversion. The vertical shaded area represents the business cycle recession dates reported by NBER.

This faster error correction is central to why FA-SOP succeeds. A sluggish reversion process leaves valuation gaps uncorrected for extended periods, making the resulting signal too weak

and too delayed to be useful at monthly or quarterly frequencies. Traditional SOP falls into this category because its reversion signal is slow-moving and only weakly translated into subsequent multiple growth. By contrast, FA-SOP compresses the duration of valuation misalignment and converts deviations in the price-earnings multiple into stronger and more timely forecasts of gm_{t+1}.

Taken together, Figures 3 and 4 point to the same conclusion. The superior performance of FA-SOP does not simply reflect a closer fitted path for the price-earnings multiple; rather, it reflects a faster and more usable error-correction process. This structural relation is difficult to uncover when one directly forecasts stationary returns, because the predictive signal operates through the deviation of the nonstationary valuation multiple from its conditional expectation and the subsequent adjustment in gm_{t+1}. More importantly, this evidence shows why valuation multiple reversion is the margin on which active modeling should be concentrated within the broader "predicting the predictable" strategy.

5.2 "Predicting the predictable" with latent macroeconomic factors

The second mechanism is measurement of the macroeconomic state. The previous subsection shows that return predictability emerges from valuation multiple reversion, but the strength of that mechanism depends critically on how accurately the conditional expectation for the price-earnings multiple is characterized. Latent macroeconomic factors play a central role in this respect. As emphasized by Jurado et al. (2015), in a data-rich environment what matters is not the raw variation of a small number of indicators, but how predictable the economy is once the forecastable component is properly removed and the relevant information set is made sufficiently rich. In our setting, factors extracted from a high-dimensional macroeconomic dataset provide a parsimonious and economically meaningful summary of the macroeconomic state, allowing the conditional expectation of pe_t to be measured more accurately than with a small number of observable predictors and thereby reinforcing the predictive content of valuation multiple reversion.

To better understand the economic content of these factors, we first examine their relationship with observable macroeconomic series. Following McCracken and Ng (2016), we regress each transformed observable series on the first k stationary macroeconomic factors. For each series i and factor $r = 1, \ldots, k$, we compute the adjusted $R_i^2(r)$, which measures the cumulative explanatory power of the first r factors. Since the factors are orthogonal and sorted by

decreasing eigenvalues, we compute their marginal contribution as

$$mR_i^2(r) = R_i^2(r) - R_i^2(r-1) \quad \text{for } r \geq 2, \qquad mR_i^2(1) = R_i^2(1).$$

We then define the average marginal explanatory power of factor r as

$$mR^2(r) = \frac{1}{N}\sum_{i=1}^{N} mR_i^2(r),$$

where N is the total number of observable series.

Table 5 summarizes the economic interpretation of the latent factors by reporting, for each factor, an economic label and one representative observable series used for interpretation. The representative series is selected from the top-ranked observable series by marginal explanatory power to best capture the common economic category of the factor. The full list of top-ranked series for each factor is reported in Online Appendix C.

Table 5: Marginal explanatory power of latent macroeconomic factors

	Monthly factors		Quarterly factors	
Factor	Label	Representative series	Label	Representative series
1	Output	INDPRO	Employment	HOANBS
2	Prices	CPIAUCSL	Industrial production	TCU
3	Interest rates	AAAFFM	Prices	CPIAUCSL
4	Interest rates	GS1	Earnings/net worth	TNWMVBSNNCBBDIx
5	Labor	UEMP15OV	Employment	CES9093000001
6	Labor	AWHMAN	Productivity/interest rates	OPHPBS
7	Stock market	S&P 500	Credit	CONSUMERx
8	Credit	CONSPI	Interest rates	GS1TB3Mx

Note: This table reports economic labels and representative observable series for the latent macroeconomic factors. The representative series is chosen from the top-ranked observable series by marginal explanatory power to best reflect the common economic content of the factor. The full marginal explanatory power table is reported in Online Appendix C.

At the monthly frequency, the eight latent factors can be interpreted as output, prices, interest rates (two distinct factors), labor (two factors), stock market conditions, and credit conditions. The corresponding observable proxies are industrial production (INDPRO), consumer price index (CPIAUCSL), Aaa–FF spread (AAAFFM), 1-year Treasury yield (GS1), long-term unemployment (UEMP15OV), average manufacturing hours (AWHMAN), the S&P 500 index, and nonrevolving consumer credit to personal income (CONSPI). At the quarterly frequency, the latent factors are somewhat less cleanly interpretable, but can still be associated with employment, industrial production, prices, earnings (two factors), credit, and interest rates. Their corresponding observable proxies are hours worked (HOANBS), capacity utilization (TCU), CPI

(CPIAUCSL), net worth to income ratio (TNWMVBSNNCBBDIx), local government employment (CES9093000001), labor productivity (OPHPBS), real consumer loans (CONSUMERx), and the term spread (GS1TB3Mx). Online Appendix C further reports the same marginal explanatory power exercise separately for NBER recession and expansion periods. The results show that the latent factors remain economically interpretable across business-cycle states and that their explanatory power is more concentrated during recessions, when the leading factors load strongly on real activity, interest-rate and credit conditions, housing/labor-market conditions, and stock-market stress.

These patterns indicate that the latent factors capture broad and economically meaningful dimensions of macroeconomic variation rather than statistical artifacts. Importantly, however, assigning observable labels to the factors does not imply that the factors can be replaced by a small set of observable proxies. Table 6 makes this point by replacing the latent factors with their selected observable counterparts. The observable proxies capture part of the information in the factors, but they do not replicate the predictive content of the full latent factor structure: when all observable proxies are included simultaneously, FA-SOP's R^2_{OOS} falls to 0.76% at the monthly frequency and 0.07% at the quarterly frequency, compared with 1.95% and 4.88%, respectively, when the full latent factor set is used. FA-Unified continues to perform poorly under the same observable-proxy substitution.

Table 6: Return predictability using observable factor proxies

		FA-Unified (observable factors)			**FA-SOP (observable factors)**		
FRED Code	Description	R^2_{OOS}	CER gain	SR gain	R^2_{OOS}	CER gain	SR gain
Panel A: Monthly frequency (January 1980–December 2022)							
INDPRO	Industrial production	−0.81	−1.40	−0.08	0.34	0.79	0.05
CPIAUCSL	Consumer price index	−0.94	−0.49	0.00	0.42	0.85	0.06
AAAFFM	Aaa–FF spread	−1.23	0.43	0.05	0.56*	1.17	0.09
GS1	1-year Treasury yield	−0.68	−0.27	−0.02	0.84**	1.43	0.10
UEMP15OV	Long-term unemployment	−0.35	1.56	0.10	0.53*	1.27	0.09
AWHMAN	Average manufacturing hours	−0.86	−1.29	−0.09	0.63**	0.90	0.06
S&P 500	S&P 500 index	−2.87	−2.28	−0.16	0.65*	1.61	0.10
CONSPI	Nonrevolving consumer credit	−0.27	−0.32	−0.02	0.60*	1.42	0.11
-	All observable factors	−7.55	−3.08	−0.22	0.76**	2.17	0.14
-	All unobservable factors	−2.47	1.46	0.09	1.95***	3.62	0.26
Panel B: Quarterly frequency (1st quarter 1980–4th quarter 2022)							
HOANBS	Hours worked	−2.38	−1.49	−0.06	1.74*	2.71	0.19
TCU	Capacity utilization	−1.61	−0.97	0.01	2.94**	3.47	0.28
CPIAUCSL	Consumer price index	−2.78	−0.91	−0.01	2.12*	2.66	0.17
TNWMVBSNNCBBDIx	Net worth to income ratio	0.88*	−0.12	0.01	2.28*	3.25	0.28
CES9093000001	Local government employment	−1.40	−0.58	0.00	2.31**	1.72	0.10
OPHPBS	Labor productivity	−2.48	−1.92	−0.09	2.02*	2.74	0.18
CONSUMERx	Real consumer loans	−2.69	−1.76	−0.08	1.90*	2.62	0.17
GS1TB3Mx	Term spread	−1.01	−1.74	−0.11	3.52**	3.51	0.27
-	All observable factors	−13.31	−1.96	−0.09	0.07	3.24	0.22
-	All unobservable factors	−12.88	−1.82	−0.10	4.88**	2.75	0.16

Note: This table reports the out-of-sample statistical and economic performance of FA-Unified and FA-SOP when the latent macroeconomic factors are replaced by their representative observable proxies. CER and Sharpe ratio gains are annualized. Asterisks denote statistical significance of the R^2_{OOS} against the historical mean benchmark using the Clark-West statistic. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

These findings clarify the role of latent factors in our framework. The factors are not

simply substitutes for observed macroeconomic variables, nor are they included only as a robustness check. They summarize common variation in the macroeconomic panel while filtering out idiosyncratic noise and reducing the instability that can arise when a small set of proxies is chosen by hand. Once the decomposition identifies the return component that should be modeled actively, the factors determine how the relevant macroeconomic state is measured.

5.3 "Predicting the predictable" beyond linear models

The analysis so far suggests that predictive gains arise when modeling effort is directed toward the return component with the strongest signal content. We next ask whether this advantage survives once the unified benchmark is made more flexible. If a nonlinear unified model could dominate the decomposition-based approach, the gains from FA-SOP would mainly reflect the limitations of linear benchmarks. If not, the decomposition itself is a more fundamental source of forecasting discipline.

To address this question, we compare two machine-learning implementations trained on the same high-dimensional macroeconomic dataset:

1. **Unified-ML (Direct Prediction):** Flexible ML algorithms forecast the aggregate stock return (r_{t+1}) directly.

2. **SOP-ML (Flexible Decomposition Framework):** The same algorithms are embedded within the SOP decomposition: dp_t and ge_t remain passively forecasted, while the ML models are used for the valuation-multiple step that generates forecasts of gm_{t+1}.

To ensure a broad evaluation, we implement 10 prominent ML methods across both paradigms: Lasso, Ridge, Elastic Net (EN), Partial Least Squares (PLS), Support Vector Regression (SVM), Random Forest (RF), Gradient Boosted Regression Trees (GBRT, implemented via XGBoost), and three neural-network architectures (NN1, NN2, NN3). Hyperparameter tuning and model specifications are detailed in Online Appendix B.

Table 7 reports the out-of-sample statistical and economic performance of the two paradigms at monthly and quarterly frequencies without transaction costs. For brevity, results with 50 basis points transaction costs are reported in Online Appendix C and lead to the same qualitative conclusions.

The difference between the two forecasting approaches is substantial. When ML algorithms are used in unified models that forecast aggregate returns directly, out-of-sample performance

Table 7: Out-of-Sample Performance of Machine Learning Methods: Unified-ML vs. SOP-ML

	Unified-ML			SOP-ML		
Method	R^2_{OOS} (%)	CER gain	SR gain	R^2_{OOS} (%)	CER gain	SR gain
Panel A: Monthly frequency (January 1980–December 2022)						
Lasso	−2.87	−1.52	−0.10	0.63*	1.49	0.12
Ridge	−0.59	−0.28	−0.01	1.50**	4.28	0.31
EN	−2.74	−1.51	−0.10	0.84**	1.79	0.13
PLS	−5.35	−1.02	−0.05	1.51***	2.70	0.18
SVM	−4.39	−1.59	−0.07	0.97**	2.87	0.20
RF	−8.17	0.27	0.02	1.81***	2.14	0.14
GBRT	−4.16	0.80	0.05	1.64**	2.19	0.17
NN1	−5.14	−1.78	−0.10	1.17**	1.74	0.12
NN2	−0.90	−1.31	−0.08	1.83***	3.97	0.30
NN3	−1.32	−1.78	−0.12	1.67***	3.41	0.25
Panel B: Quarterly frequency (1st quarter 1980–4th quarter 2022)						
Lasso	−7.57	−0.52	−0.02	−1.43	0.87	0.04
Ridge	−2.44	−0.67	−0.03	4.99**	4.71	0.34
EN	−7.23	−1.01	−0.05	−1.35	0.99	0.05
PLS	−11.29	−2.61	−0.12	6.59***	4.77	0.31
SVM	0.03	0.20	0.05	7.51***	5.21	0.35
RF	−4.34	−1.30	−0.06	3.82**	4.18	0.31
GBRT	−2.34	1.16	0.07	3.10**	3.38	0.25
NN1	−21.12	−0.72	−0.03	−7.19	1.30	0.07
NN2	−1.13	−0.15	0.00	3.84**	1.27	0.07
NN3	−0.31	0.35	0.03	2.29**	3.08	0.19

Note: This table presents a side-by-side comparison of the out-of-sample statistical and economic performance of various machine learning algorithms under two different frameworks, without incorporating transaction costs. The "Unified-ML" models (left) forecast the aggregate stock return directly using the high-dimensional macroeconomic dataset. The "SOP-ML" models (right) embed the same set of machine learning algorithms within our FA-SOP framework to forecast the price-earnings multiple growth (gm). Asterisks denote the significance of the R^2_{OOS} against the historical mean benchmark, evaluated using the Clark-West (2007) statistic. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

is generally poor. At the monthly frequency, all Unified-ML models produce negative R^2_{OOS} values, ranging from −0.59% for Ridge to −8.17% for Random Forest. The quarterly results are similarly weak, with R^2_{OOS} values of −21.12% for NN1 and −11.29% for PLS. These direct ML forecasts also generate negative CER and Sharpe ratio gains, indicating that the weak statistical performance carries over to economic value.

This widespread underperformance highlights the econometric challenges inherent in pure aggregate time-series forecasting. This evidence should be distinguished from the positive portfolio-level predictability documented by Gu et al. (2020). Their S&P 500 forecasts are constructed bottom-up by aggregating machine-learning forecasts of individual stock returns, so the portfolio result benefits from a large stock-level panel and from averaging out idiosyncratic return noise. Their out-of-sample R^2 is also evaluated against a zero benchmark rather than the more demanding historical-mean benchmark (Campbell and Thompson, 2008) used

here. By contrast, our Unified-ML benchmark operates in a pure aggregate macro-forecasting environment. As emphasized by Israel et al. (2020), such settings are characterized by a low signal-to-noise ratio and a challenging ratio of predictors to time-series observations. Under these conditions, flexible nonlinear algorithms are prone to overfitting transitory market noise rather than capturing stable expected-return variation.

The same algorithms perform much better once they are placed inside the decomposition framework. At the monthly frequency, SOP-ML produces positive and statistically significant R^2_{OOS} values across all ML methods, reaching 1.83% for SOP-NN2, with annualized CER gains up to 4.28% and Sharpe-ratio gains up to 0.31 for SOP-Ridge. At the quarterly frequency, SOP-SVM, SOP-PLS, and SOP-Ridge produce statistically significant R^2_{OOS} values of 7.51%, 6.59%, and 4.99%, respectively, along with CER gains above 5%. These gains remain robust when 50 basis points transaction costs are introduced. Our framework therefore achieves predictive performance comparable to data-intensive bottom-up aggregation approaches such as Gu et al. (2020), while relying on a more parsimonious set of aggregate macroeconomic predictors.

Taken together, the ML results reinforce the paper's central interpretation. The advantage of FA-SOP is not limited to the baseline linear factor specification, nor does it come simply from using more flexible algorithms. Predictive signals are recovered more reliably when flexible methods are directed toward the valuation component rather than toward aggregate returns, where components with different signal content are pooled together.

6 Simulation analysis

The empirical evidence points to a target-design mechanism, but realized returns do not reveal the true expected return. This section therefore uses a controlled environment in which the expected return is known. We build a factor-augmented present-value model in which macroeconomic information may be weakly related to one-period realized returns but more visible through valuation multiples. We estimate the structural parameters using historical annual data and use the resulting data-generating process (DGP) to simulate the joint dynamics of returns and fundamentals. The simulation asks whether the competing forecasting methods recover the true expected return when the signal is organized as in the present-value relation.

6.1 A factor-augmented present-value model

The simulation builds on the log-linear present-value relation introduced in Section 2.2. Equation (7) is the channel through which the simulation links macroeconomic information to valuation: the price-earnings multiple reflects expected future returns, earnings growth, and payout policy.

To introduce macroeconomic information, let $\mathbf{X}_t$ denote a high-dimensional macroeconomic dataset and let $\mathbf{F}_t$ denote a small number of latent factors that summarize its common variation:

$$\mathbf{X}_t = \mathbf{\Lambda}\mathbf{F}_t + \epsilon_t. \tag{23}$$

The factors capture the aggregate state of the economy and enter investors' expectations about future returns, earnings growth, and payout policy. We specify the expectation processes as

$$\mu_{t+1} = \mu_0 + \delta_1(\mu_t - \mu_0) + \boldsymbol{\delta}_2'\mathbf{F}_t + \varepsilon^{\mu}_{t+1}, \tag{24}$$

$$g_{t+1} = g_0 + \gamma_1(g_t - g_0) + \boldsymbol{\gamma}_2'\mathbf{F}_t + \varepsilon^{g}_{t+1}, \tag{25}$$

$$\phi_{t+1} = \phi_0 + \lambda_1(\phi_t - \phi_0) + \boldsymbol{\lambda}_2'\mathbf{F}_t + \varepsilon^{\phi}_{t+1}, \tag{26}$$

$$\mathbf{F}_{t+1} = \mathbf{\Theta}\mathbf{F}_t + \boldsymbol{\varepsilon}^{F}_{t+1}, \tag{27}$$

where

$$\mu_t = E_t(r_{t+1}), \qquad g_t = E_t(\Delta e_{t+1}), \qquad \phi_t = E_t(de_{t+1}).$$

Here, μ_0, g_0, and ϕ_0 denote the unconditional means of expected returns, expected earnings growth, and expected payout policy, respectively. The realized earnings growth and payout ratio are given by

$$\Delta e_{t+1} = g_t + \varepsilon^{e}_{t+1}, \qquad de_{t+1} = \phi_t + \varepsilon^{de}_{t+1}. \tag{28}$$

Taking conditional expectations in equation (7) and substituting the state dynamics in equations (24)–(27), the log price-earnings multiple can be expressed as

$$pe_t = A - B_1\hat{\mu}_t + B_3\hat{g}_t + B_5\hat{\phi}_t + \mathbf{B}_7'\mathbf{F}_t, \tag{29}$$

where $\hat{\mu}_t = \mu_t - \mu_0$, $\hat{g}_t = g_t - g_0$, and $\hat{\phi}_t = \phi_t - \phi_0$. The coefficients A, B_1, B_3, B_5, and $\mathbf{B}_7$ are functions of the persistence parameters and factor loadings in the expectation processes. The

detailed derivation is provided in Online Appendix F.

Equation (29) is the key implication for the simulation. It makes the price-earnings multiple depend on expected returns, expected earnings growth, expected payout policy, and macroeconomic state variables, allowing us to evaluate whether factor-based measurement of the conditional valuation level improves forecasts relative to low-dimensional SOP-reversion and unified return prediction.

6.2 Estimation of the factor-augmented present-value model

For parsimony and to avoid over-parameterizing the state-space system, we specify a single-factor structure ($k = 1$), utilizing the first principal component extracted from our high-dimensional dataset to capture the dominant source of aggregate macroeconomic variation. We cast the model into a compact state-space representation and estimate its structural parameters via maximum likelihood, utilizing the Kalman filter on annual U.S. stock market data from 1960 to 2022. The complete state-space derivation and the Kalman filter algorithm are detailed in Online Appendix G.

The maximum likelihood estimates (reported in Online Appendix G) provide an economically coherent calibration that is consistent with the established asset-pricing literature (e.g., Campbell and Cochrane, 1999; Pástor and Stambaugh, 2009). A useful feature of the estimates is that the macroeconomic factor ($\mathbf{F}_t$) has only a weak and statistically insignificant direct loading on single-period expected returns. At the same time, the present-value relation allows even modest macroeconomic effects on expected returns, earnings growth, or payout policy to appear in valuation multiples through expectations over many future horizons. The estimated model therefore reproduces the empirical tension that motivates FA-SOP: the direct short-horizon return signal is weak, while the valuation multiple contains more persistent macroeconomic state dependence.

With these estimated DGP parameters, we proceed to Monte Carlo simulations that benchmark the out-of-sample performance of the competing forecasting methods.

6.3 Simulation-based comparison of forecasting methods

To construct our simulated economies, we generate 10,000 independent replications of 100-year annual time series based on the fully parameterized present-value model. In each replication, we draw normally distributed structural shocks ($\varepsilon_t^{\mu}, \varepsilon_t^{g}, \varepsilon_t^{\phi}, \varepsilon_t^{e}, \varepsilon_t^{de}, \varepsilon_t^{\mathbf{F}}$) from the estimated

covariance matrix. Driven by these shocks, we iteratively generate the latent expectations (μ_t, g_t, ϕ_t), the exogenous factor $(\mathbf{F}_t)$, and the observable fundamentals $(r_t, pe_t, \Delta e_t, de_t)$ following the present-value identity in equation (7) and the system dynamics in equations (24)–(28).

Because the true latent factor $\mathbf{F}_t$ is inherently unobservable in practice, we generate a high-dimensional observable dataset $\mathbf{X}_{it}$ structurally spanned by this factor:

$$\mathbf{X}_{it} = \lambda_i \mathbf{F}_t + \epsilon_{it},$$

where the factor loadings $\lambda_i \sim$ i.i.d. Uniform$[0, 1]$ and the idiosyncratic noises $\epsilon_{it} \sim$ i.i.d. $\mathcal{N}(0, 1)$ for $i = 1, \ldots, N$. We set $N = 227$ to precisely match the cross-sectional dimensionality of the empirical FRED-QD macroeconomic dataset.

To implement the SOP-based methodologies within the simulation, we reconstruct the three mechanical components of stock returns as follows:

$$gm_t = \Delta pe_t, \quad ge_t = \Delta e_t, \quad dp_t = \log\left(1 + \exp(de_t - pe_t)\right).$$

We conduct an expanding-window forecasting exercise with an initial training window size of 20 years. This initialization period mirrors the empirical design. In each simulated economy, we evaluate seven forecasting approaches that reproduce the main empirical comparisons: passive benchmarks, direct return-prediction models, low-dimensional SOP reversion, and factor-augmented SOP reversion.

1. **FA-SOP**: Our proposed methodology. It forecasts valuation components (gm_{t+1}) using a conditional error-correction mechanism, where the conditional expectation of pe_t is fitted using the first principal component $\hat{\mathbf{F}}_t$ extracted from the high-dimensional dataset $\mathbf{X}_{it}$.

2. **FA-SOP with True Factor**: A diagnostic version of FA-SOP that replaces the estimated principal component $\widehat{\mathbf{F}}_t$ with the true latent factor $\mathbf{F}_t$ from the simulation DGP. All forecasting regressions are still estimated recursively, as in FA-SOP. This specification therefore isolates factor-extraction error; it does not use the true expected return or the true structural coefficients.

3. **Simple SOP**: The baseline Sum-of-the-Parts method that passively assumes $\widehat{gm}_{t+1} = 0$.

4. **SOP with Multiple Reversion**: A traditional version of SOP where we use the sim-

ulated dividend-price ratio (dp_t) as a univariate observable predictor in the multiple reversion step, consistent with common literature practices (e.g., Welch and Goyal, 2008; Campbell and Thompson, 2008).

5. **FA-Unified**: A direct predictive regression of aggregate returns r_{t+1} on the estimated first macroeconomic factor $\widehat{\mathbf{F}}_t$, using the same simulated macroeconomic information as FA-SOP but without the decomposition.

6. **Univariate Regression**: A standard univariate predictive regression using the simulated dividend-price ratio (dp_t) as the sole predictor for aggregate returns.

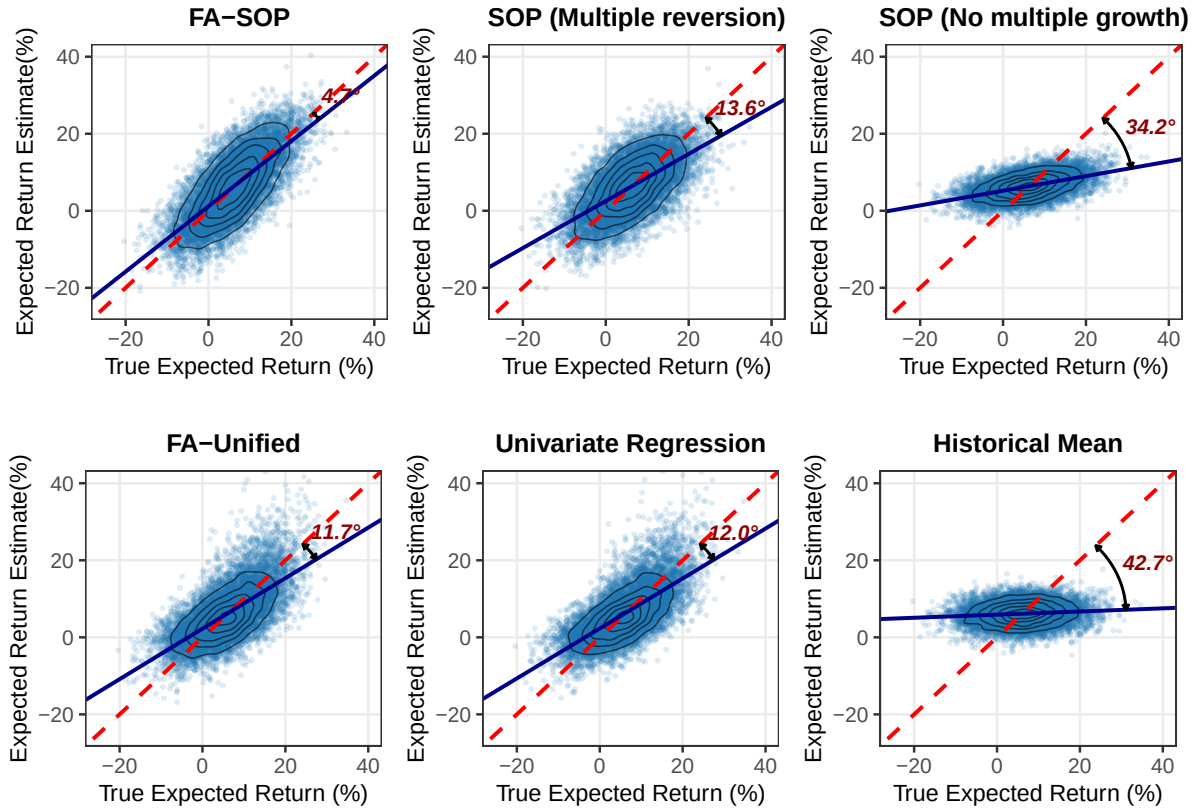
7. **Historical Mean (HM)**: A passive benchmark model that assumes constant expected returns, computed as the simple historical average over the expanding window.

Figure 5 compares the out-of-sample forecasting performance of the competing methods. For each simulated economy, we plot the estimated expected returns (y-axis) against the true expected returns implied by the data-generating process (x-axis) at the end of the out-of-sample period. To mitigate visual overplotting and reveal the distributional structure of the forecasts, we superimpose 2D kernel density contours (solid black lines) and the empirical ordinary least squares (OLS) best-fit line (solid dark blue line) over the theoretical 45-degree benchmark of perfect foresight (dashed red line). The annotated angle summarizes the degree of attenuation bias, measured by the deviation of the empirical fit from a one-to-one mapping.

As shown in the figure, the Historical Mean (HM) and the Simple SOP (No multiple growth) produce almost horizontal fit lines (42.7° and 34.2°, respectively) and horizontally flat density contours. While these passive approaches effectively suppress estimation variance, they suffer from severe predictive bias by failing to capture the dynamic, macro-driven fluctuations in true expected returns.

Direct predictive methods, including the Univariate Regression and FA-Unified, attempt to capture these dynamics but face a bias-variance tradeoff. Their density contours are more dispersed, consistent with higher forecast variance in finite samples. Their empirical fit lines are also flattened relative to the 45-degree benchmark (12.0° and 11.7°), indicating attenuation in the estimated expected-return signal. The SOP with multiple reversion offers a structural compromise by focusing on the valuation multiple. Although it still exhibits attenuation bias (13.6°), its density contours are tighter than those of FA-Unified, generating a variance-reduction benefit that partly offsets its larger bias.

Figure 5: Monte Carlo simulation: Estimated expected returns versus true expected returns



Note: This figure presents scatter plots of the estimated expected returns versus the true expected returns at the end of 10,000 simulated samples, each consisting of 100 years of data. The simulations are generated from the factor-augmented present-value model, producing time series of returns, price-earnings growth, earnings growth, and the dividend-price ratio. Each subplot corresponds to a different forecasting method. To illustrate the underlying joint distribution, 2D kernel density contours (solid black lines) are superimposed on the scatter points. The dashed red line represents the 45-degree theoretical benchmark indicating perfect foresight. The solid dark blue line denotes the empirical ordinary least squares (OLS) best-fit line of the estimates against the true expectations. The annotated angle quantifies the geometric deviation of the empirical fit line from the 45-degree benchmark.

By contrast, the FA-SOP framework tracks the target relation more closely, mainly because it reduces attenuation bias without materially increasing forecast variance relative to direct predictive regressions. Its density contours are more aligned with the 45-degree line, and the empirical fit line is closer to the theoretical benchmark (4.7°). This pattern indicates that using macroeconomic factors to measure the conditional valuation level helps the forecast covary more closely with the true time-varying expected return. Simulations using the true, unobserved macroeconomic factor, omitted for brevity, yield very similar scatter patterns and attenuation angles to estimated FA-SOP. This suggests that PCA estimation error is not the main driver of the simulation results.

While the scatter plots and density contours visually reveal how different methods trade off attenuation bias against forecast variance, a formal statistical evaluation is required to quantify

overall predictive accuracy. To this end, we compute the root mean squared error (RMSE) between the out-of-sample forecasts and the true expected returns across all simulations. Panel A of Table 8 reports the cross-simulation average RMSE and its distribution percentiles over the 10,000 Monte Carlo replications.

Table 8: Monte Carlo simulation: Mean squared error of return forecasts.

	Historical mean	Univariate regression	FA-Unified regression	SOP no multiple growth	SOP multiple reversion	FA-SOP multiple reversion	(True factor) FA-SOP multiple reversion
Panel A: Distribution of the root mean squared error (×100)							
Mean	7.342	6.259	6.711	6.104	6.067	5.699	5.699
10th percentile	5.537	4.812	5.132	4.694	4.858	4.591	4.591
25th percentile	6.246	5.329	5.718	5.244	5.352	5.055	5.055
Median	7.173	6.036	6.498	5.944	5.976	5.606	5.606
75th percentile	8.250	6.949	7.501	6.808	6.684	6.257	6.257
90th percentile	9.398	7.989	8.569	7.746	7.400	6.912	6.913
Panel B: Mean squared forecast error decomposition (×1000)							
Squared bias of forecasts	0.001	0.006	0.006	0.007	0.003	0.006	0.006
Variance of forecasts	1.343	5.647	6.449	1.523	4.465	5.759	5.759
Variance of true expected returns	4.930	4.930	4.930	4.930	4.930	4.930	4.930
−2×covariance of forecasts and true expected returns	−0.719	−6.545	−6.748	−2.632	−5.663	−7.405	−7.405
MSFE	5.625	4.089	4.695	3.875	3.782	3.332	3.332
MSFE minus Variance of true expected returns	0.695	−0.841	−0.235	−1.055	−1.149	−1.599	−1.599

Note: This table presents the mean squared error results for return forecasts from various methods in our simulation. The simulation generates 10,000 samples of 100 years of returns, price-earnings growth, earnings growth, and the dividend-price ratio using the present-value model with macroeconomic factors. Panel A reports the mean, median, and other percentiles (across simulations) of the root mean squared errors (RMSE) of each method. Panel B reports each component of the mean squared forecast error (MSFE) decomposition. The final column is FA-SOP multiple reversion with the true factor: it replaces the estimated principal component used by FA-SOP with the true latent factor from the simulation DGP, while keeping the same recursively estimated forecasting regressions. It is included only to diagnose factor-extraction error. The squared bias of forecasts is given by $\frac{1}{T-t_0}\sum_{t=t_0}^{T}\left[\mathrm{E}\left(\hat{\mu}_t-\mu_t\right)\right]^2$. The variance of forecasts is given by $\frac{1}{T-t_0}\sum_{t=t_0}^{T}\mathrm{Var}\left(\hat{\mu}_t\right)$. The variance of true expected returns is given by $\frac{1}{T-t_0}\sum_{t=t_0}^{T}\mathrm{Var}\left(\mu_t\right)$. The covariance of forecasts and true expected returns is given by $\frac{1}{T-t_0}\sum_{t=t_0}^{T}\mathrm{Cov}\left(\hat{\mu}_t,\mu_t\right)$.

The FA-SOP method achieves the lowest average RMSE at 5.699%. This RMSE is lower than all feasible alternatives: 6.104% for Simple SOP (no multiple growth), 6.067% for SOP with multiple reversion, 6.711% for FA-Unified, 6.259% for univariate regression, and 7.342% for the historical mean. The true-factor diagnostic delivers essentially the same RMSE as feasible FA-SOP, indicating that factor-extraction error is negligible in this simulation. The main comparison is therefore not between FA-SOP and its true-factor variant, but between decomposition-based valuation forecasting and direct return forecasting under the same simulated macroeconomic environment. These differences persist across nearly all percentiles of the RMSE distribution. For instance, at the median, FA-SOP remains the most accurate with an RMSE of 5.606%, whereas the historical mean and FA-Unified exhibit much higher errors of 7.173% and 6.498%, respectively.

To better understand the differences in mean squared forecast error (MSFE) across methods, we decompose the expected MSFE (averaged across simulations) of each estimator of

expected returns as follows:

$$\frac{1}{T-t_0}\sum_{t=t_0}^{T}\mathbb{E}\left[(\hat{\mu}_t-\mu_t)^2\right]=\underbrace{\frac{1}{T-t_0}\sum_{t=t_0}^{T}\left[\mathbb{E}(\hat{\mu}_t-\mu_t)\right]^2}_{\text{Squared bias}}+\underbrace{\frac{1}{T-t_0}\sum_{t=t_0}^{T}\text{Var}(\hat{\mu}_t)}_{\text{Forecast variance}}$$
$$+\underbrace{\frac{1}{T-t_0}\sum_{t=t_0}^{T}\text{Var}(\mu_t)}_{\text{True variance}}-\underbrace{\frac{2}{T-t_0}\sum_{t=t_0}^{T}\text{Cov}(\hat{\mu}_t,\mu_t)}_{\text{Covariance term}}, \tag{30}$$

where t_0 is the initial forecast time, and T is the end of the evaluation window. The first term captures the squared bias of the forecasts of expected returns, the second term is the variance of the forecasted expected returns, and the third term is the variance of the true expected returns (which is the same for all methods). The last term is the covariance between forecasts and actual expected returns. Panel B of Table 8 reports the results of this decomposition.

To better highlight the relative performance, we report MSFE values net of the true expected return variance in the last row of Table 8. The results show that FA-SOP achieves the lowest adjusted MSFE, with a value of -1.599. In comparison, while the Simple SOP and SOP with multiple reversion also yield negative adjusted MSFE values (-1.055 and -1.149, respectively), they are less effective than FA-SOP. For the remaining methods, such as the historical mean (0.695) and the FA-Unified (-0.235), the values indicate significantly inferior performance compared to the FA-SOP benchmark.

Within the structure of the present-value model, no method exhibits a substantial disadvantage in terms of the squared unconditional bias (centering bias), as all values remain close to zero (e.g., 0.006 for FA-SOP).[12] The main driver of performance lies in the second moments. FA-SOP has the largest covariance (in absolute terms) with true expected returns (-7.405) among the methods considered, which reduces its overall MSFE. In contrast, the covariance components for other methods, such as the historical mean (-0.719) and Simple SOP (-2.632), are weaker. Although FA-Unified also attains a relatively high covariance (-6.748), its higher forecast variance (6.449) offsets much of this benefit, leading to inferior overall performance.

Overall, the simulation supports the mechanism emphasized in the empirical analysis. When the true expected return is observed by the researcher, direct aggregate-return forecasts

[12] We emphasize the formal econometric distinction between this unconditional forecast bias (centering bias) and the conditional attenuation bias (slope bias) analyzed in Figure 5. The unconditional bias captures a systematic baseline shift (i.e., whether the forecasts over- or under-predict the true returns on average), which is naturally small across all methods. In contrast, the attenuation bias reflects a conditional compression in the amplitude of the forecasts, manifesting geometrically as a flattened slope rather than a simple vertical shift.

exhibit high forecast variance and attenuated expected-return signals. Targeting macro-implied valuation adjustment allows FA-SOP to track variation in true expected returns more closely. The simulation therefore closes the evidence chain: the empirical gains are consistent with a known-DGP environment in which macroeconomic information is difficult to use through realized returns but more recoverable through valuation adjustment.

7 Robustness Checks

Additional analyses support the main target-design interpretation. First, we examine forecast performance separately across NBER recessions and expansions. FA-SOP remains statistically strong in both states, but its gains are especially pronounced during recessions. This pattern is consistent with the factor-interpretation evidence in Section 5: macroeconomic comovement is more concentrated in downturns, and valuation multiples are more likely to adjust to macro-financial conditions in these periods. The business-cycle results therefore reinforce the view that FA-SOP is using macroeconomic information through valuation adjustment rather than simply adding a broad information set to a return regression.

Second, we vary the predictor set. When FA-SOP is estimated with one latent factor at a time, the results remain positive but are weaker than those from the full factor specification. This indicates that no single factor fully captures the relevant macroeconomic state. Conversely, expanding the observable-predictor SOP specification into a kitchen-sink regression does not improve performance. The direct predictive regression with many observable predictors performs poorly, and the kitchen-sink SOP specification also underperforms the factor-based FA-SOP benchmark.

Taken together, these robustness exercises show that the baseline gains are not driven merely by using more predictors. The results depend on two linked choices: summarizing macroeconomic information through latent factors and directing that information toward the valuation-adjustment component identified by the return decomposition. Full results for the business-cycle exercise, alternative predictor-set specifications, observable-factor substitutions, and additional factor interpretation are reported in Online Appendix C.

8 Conclusion

We revisit return predictability from the perspective of "predicting the predictable." The central lesson is that the weak performance of aggregate return forecasts need not imply that macroeconomic information is irrelevant for expected returns. It may instead indicate that the forecasting target is not organized in the most informative way. Aggregate returns combine components with different signal-to-noise ratios, and direct return regressions can obscure signals that are more visible in valuation adjustment.

Motivated by this view, we propose the factor-augmented sum-of-the-parts (FA-SOP) framework. FA-SOP retains parsimonious forecasts for lower-signal components and uses high-dimensional macroeconomic information to forecast valuation-multiple growth. Applied to S&P 500 returns from 1960 to 2022, FA-SOP delivers robust out-of-sample gains relative to the historical mean, univariate regressions, FA-Unified regressions, and traditional SOP specifications. The key comparison is not simply that FA-SOP performs better, but that the same underlying macroeconomic panel performs poorly when used to forecast aggregate returns directly and becomes useful when routed through valuation adjustment.

The analysis further shows that the sources of predictability are structural rather than purely algorithmic. Valuation multiple reversion provides the main forecasting margin; latent macroeconomic factors help extract the relevant common forecastable signal from a data-rich information set; and the machine learning extension confirms that even highly flexible algorithms perform poorly when they forecast aggregate returns directly, but improve substantially when organized around the predictable component. A present-value simulation with known expected returns supports the same interpretation: decomposition helps recover time-varying expected returns by directing information to the component where the signal is concentrated.

Overall, our results suggest that progress in return forecasting may come less from increasing model complexity per se than from imposing economic discipline on the forecasting target. A reliable time-varying estimate of expected market returns can also serve as an input for broader applications, including portfolio allocation, valuation, cost-of-capital estimation, and conditional asset-pricing analysis. In particular, more reliable estimates of the conditional expected market premium can improve CAPM-implied mappings from market risk premia to industry expected returns, thereby providing useful inputs for cost-of-equity measurement and conditional-CAPM diagnostics. More generally, the same logic may be useful whenever noisy

aggregate financial outcomes can be decomposed into components with different degrees of predictability. The FA-SOP framework provides one implementation of this idea, but the underlying principle is broader: meaningful signals in data-rich environments are more likely to emerge when prediction is organized around the parts of returns that are economically and statistically most predictable.

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